



Ultima 10

Workforce Data Collection Terminal



Installation Guide

This guide provides installation instructions only. For information regarding actual operation and configuration of the **Ultima 10** Time & Attendance terminal, refer to the user manual.



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Standard Modules

Wi-Fi

4G

Relay

Internal Optional Modules

PoE+

Backup battery

Cameras:

FAM33 Facial Recognition/2MP Camera

Card modules:

RFID/MultiCard reader

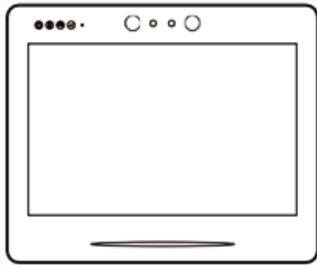
External Optional Modules

Fingerprint Scanners:

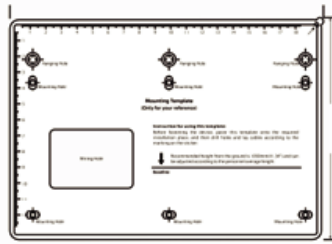
Green-light Fingerprint / White-light Fingerprint

Card Readers:

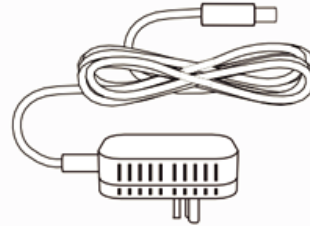
Magnetic/Barcode



Terminal



Mounting Template



Power Supply



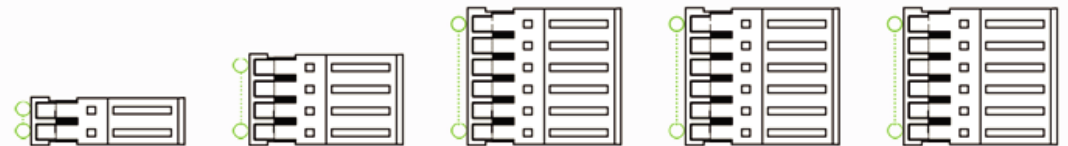
Network Cable



Keys(2)



Screws & Anchors



Relay Connector 2-pin (1x), 4-pin (1x), 6-pin (3x) soldered on motherboard (Standard)

The contents of the package may vary depending on the modules that you choose.

Pre-Installation

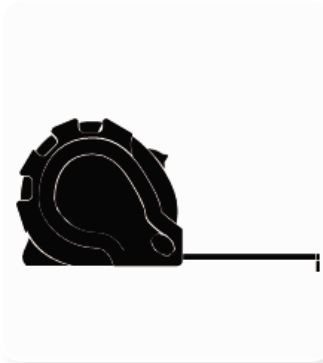


Important

To avoid damaging the main board, always wear an anti-static wrist strap when touching the main board in any way, including connecting and disconnecting wires.

- Do not remove the cover assembly from its anti-static packaging until you are ready to install the device
- Save the paper mounting template and any other printed documents

Note: Because of the small size of certain screws, we recommend that you use small screwdrivers with magnetic tips.



Measuring Tape



Level



Wire Cutter



Screwdrivers



Drywall Knife

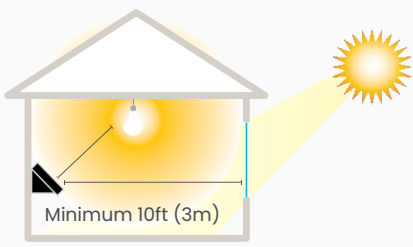


Power Drill



Location

- Ultima 10 is designed for any internal environment. In rare cases, user sites may contain agents in nature that can corrode the electronics assemblies in these devices. In the event of this situation, users must move the devices to a suitable environment
- Install the Ultima 10 in an area where the screen is not exposed to direct sunlight or other high-intensity lighting that could make the screen difficult to read

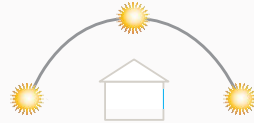


Minimum 10ft (3m)

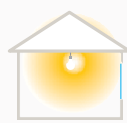
Lux References Guide



10 LUX



1200+ LUX

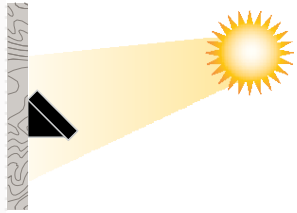


0-800 LUX (Recommended)

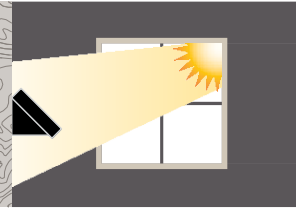
Recommended Locations

The device should only be installed indoors. The illuminance of the external light source is 0-800LUX. Direct sunlight should be avoided. Install the device at least 10 feet (3m) away from windows, and 6.5 feet (2m) away from light fixtures.

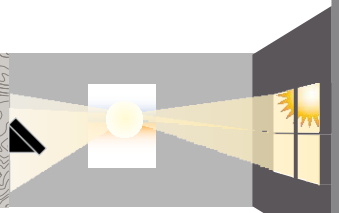
Avoid these situations



Direct sunlight outdoors



Direct sunlight through windows



Sunlight reflection off an object



Too close to light fixtures

Avoid direct sunlight, indirect sunlight reflected off mirrors, or bright artificial light.



Warning Relay is rated for **12V only** and **should not exceed the recommended voltage**. If connecting a higher voltage device some extra components maybe necessary. Please consult ZKTeco WFM for further guidance before connecting to higher voltages.



Required Personnel

Depending on the modules that you purchased and the implementation planned for your site, you will require the following:

- A licensed electrician to install or move AC outlets and power cables
- A qualified person to run Ethernet cable

One or more qualified installers to assemble the device and mount it on the wall (a second person may be necessary to hold the cover while the other person attaches components to the cover and base).



Mounting Template

The packaging box includes a paper mounting template that you use to mark the wall with the following measurements and indicators:

- Where to screw the base of the unit into the wall
- Where cables enter the unit through access holes in the wall



Power Sources and Connections

Requirements:

- The AC power source must be grounded 100 to 240V AC, 50/60Hz input voltage
- A licensed electrician must ensure that low-voltage wiring is routed securely away from high-voltage wiring

Power source options:

The following power supply methods are available:

- Mounted near an AC outlet, with power supply plugged in
- Power-over-Ethernet+ (PoE+)
 - Warning** Power up your device with AC adapter or Power-over-Ethernet+ (PoE+). Do not connect simultaneously (It may damage the device)
- Backup battery



Ethernet Connection

- Location - Plan to install the Ultima 10 in a location that allows an easy and secure connection to an external network (for example, using the Ethernet cable)
- All cables must be in place before installing the Ultima 10



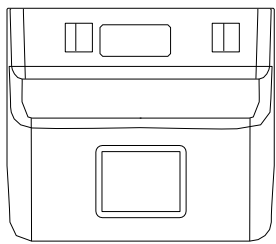
If you plan to install certain optional modules to the Ultima 10, follow the step-by-step instructions in this guide.

Note the following:

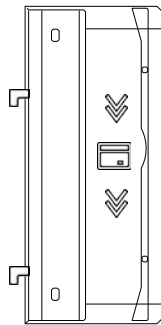
- The optional modules have components and cables that must be attached to the Ultima 10 in a certain order during specific stages of the installation
- You can only choose one between the barcode and magnetic modules.

The modules combination can be as follows:

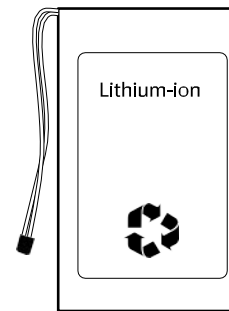
- Base+ Fingerprint & RFID module
- Base+ RFID module
- Base+ Facial Recognition module
- Base+ Barcode/Magnetic module
- Internal PoE+ Module



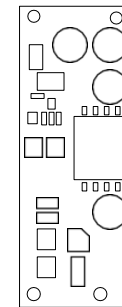
Fingerprint module



Card Reader



Battery



PoE+ module



Determine the power source for your site

Before you start the installation, plan with your electrician (if required for the power option that you choose) how the Ultima 10 will connect to the power at your site.

Choose from the following options and review the associated notes.

Power options

- Install near an AC outlet
Select a location where the distance from the wiring hole of the Ultima 10 is not more than 5 feet (1.5 meters) from the AC outlet
Note: Installing the device near an AC outlet does not protect against the AC power cord from being deliberately or inadvertently unplugged from the outlet.
- Power-over-Ethernet+ (PoE+)
The device has a PoE+ module in it. You can supply power to the device by connecting a network cable to the PoE+ switch
- Backup battery
When the power is off or in emergency, the backup battery can supply power to the device

Notes :

- A licensed electrician must install the AC outlet and associated wiring before you install the Ultima 10
- Do not connect the Ultima 10 power line to circuits that are being used for electrical devices that draw large amounts of power, such as air conditioning units, electrical motors, and compressors
- Avoid running a communications cable near devices that interfere with data transmission

Installation

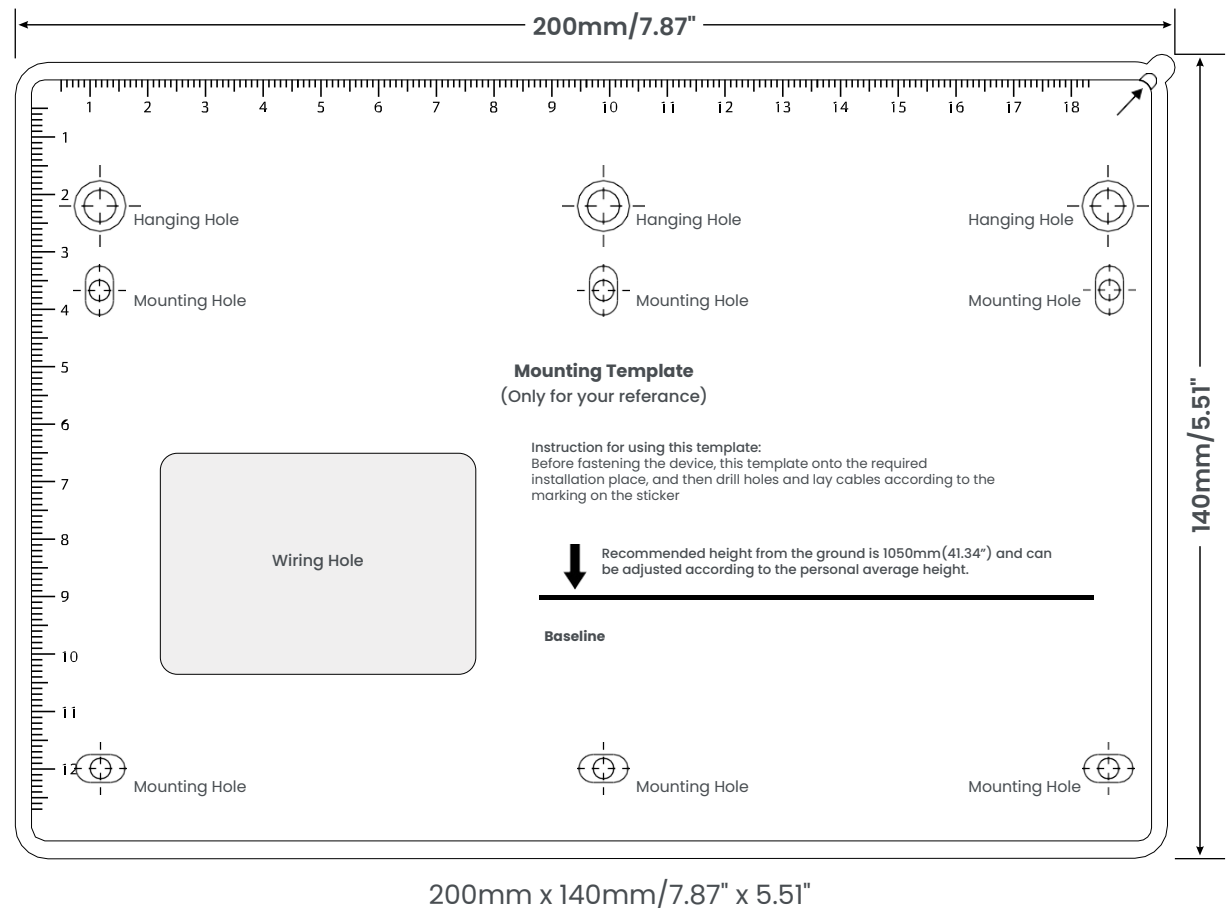


Mark insertion points for the base

The mounting template is printed on a sheet of paper that is included in the package. Use that template to mark the areas of the wall where you plan to install the Ultima 10.

Mark the wall

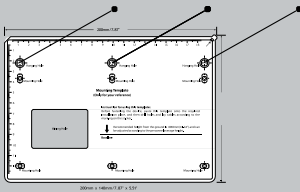
1. Tape the mounting template at the optimal height and location for the Ultima 10
 - a. The baseline is recommended 1.2m above the floor and can be adjusted to the actual situation
 - b. Use a level to ensure that the top of the template is horizontal to the floor
2. Mark the location of the mounting screw holes
3. If you plan to run an Ethernet cable through the wall to enter the Ultima 10 from the back or through the top, use the template to mark the area of entry
4. If you are installing the Ultima 10 near an AC outlet, be sure the outlet is within 5 feet (1.5 meters) of the unit
5. After you finish marking all screw holes and wiring hole, remove the mounting template from the wall



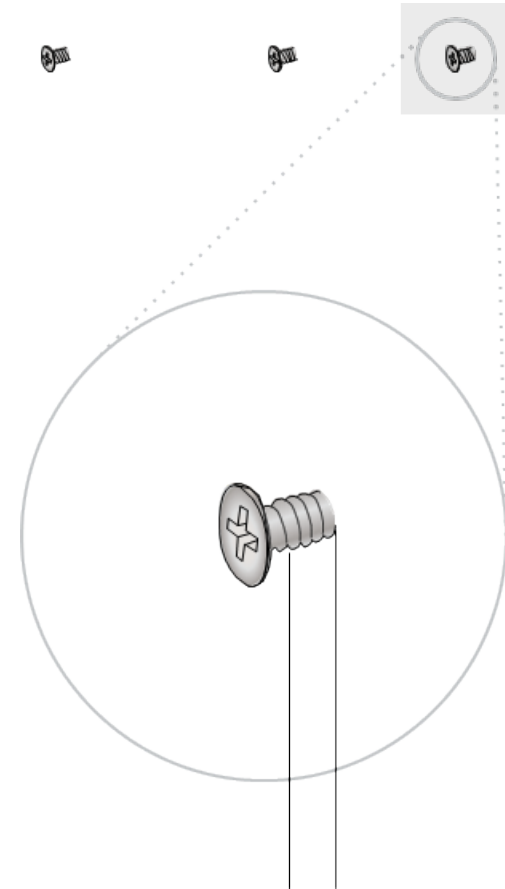
Drilling holes and fixing plastic anchors and upper screws



Drill 1/4 inch holes and insert plastic anchors flush with the wall surface

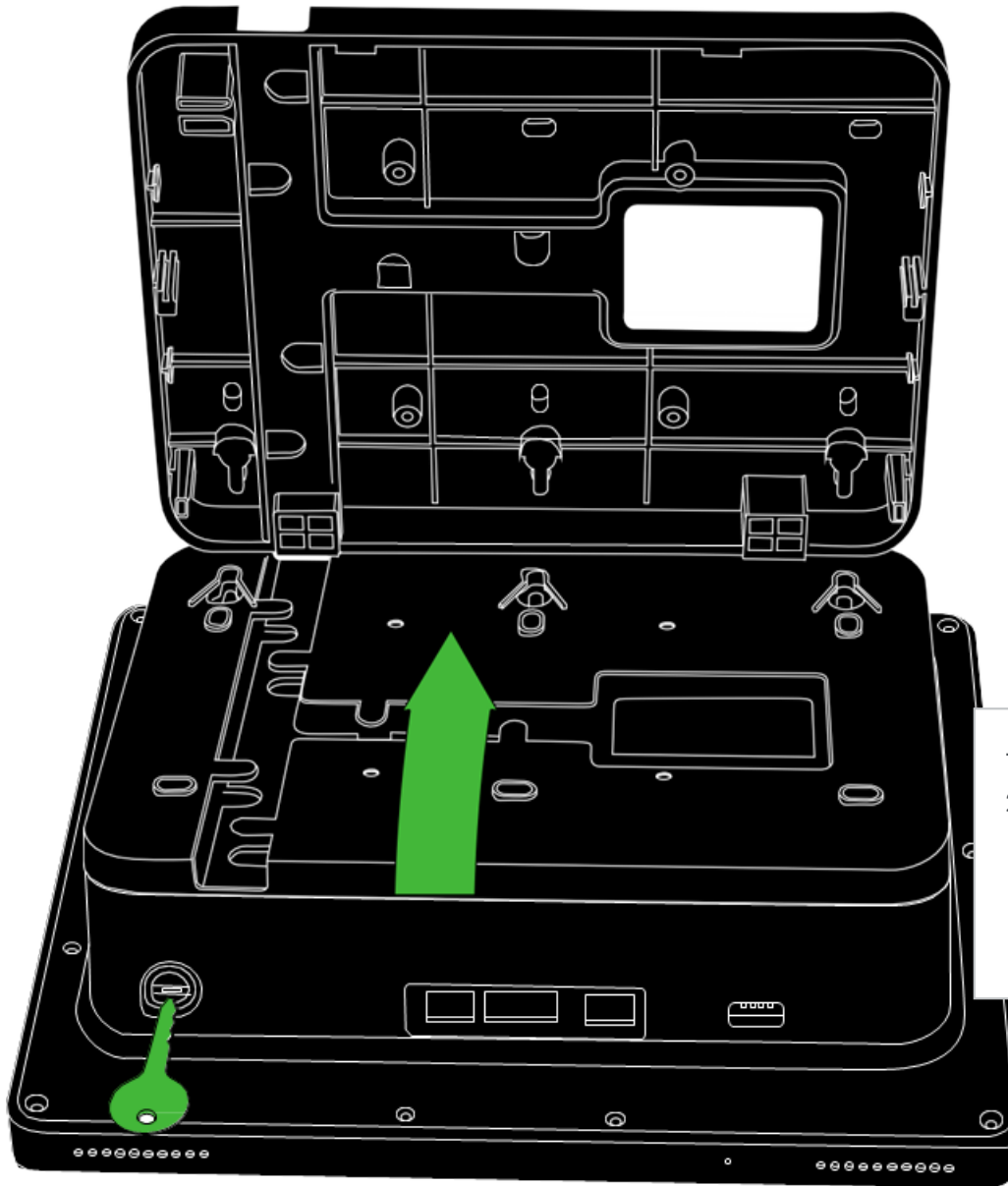


Fix the upper 3 screws (base only installation) on the wall



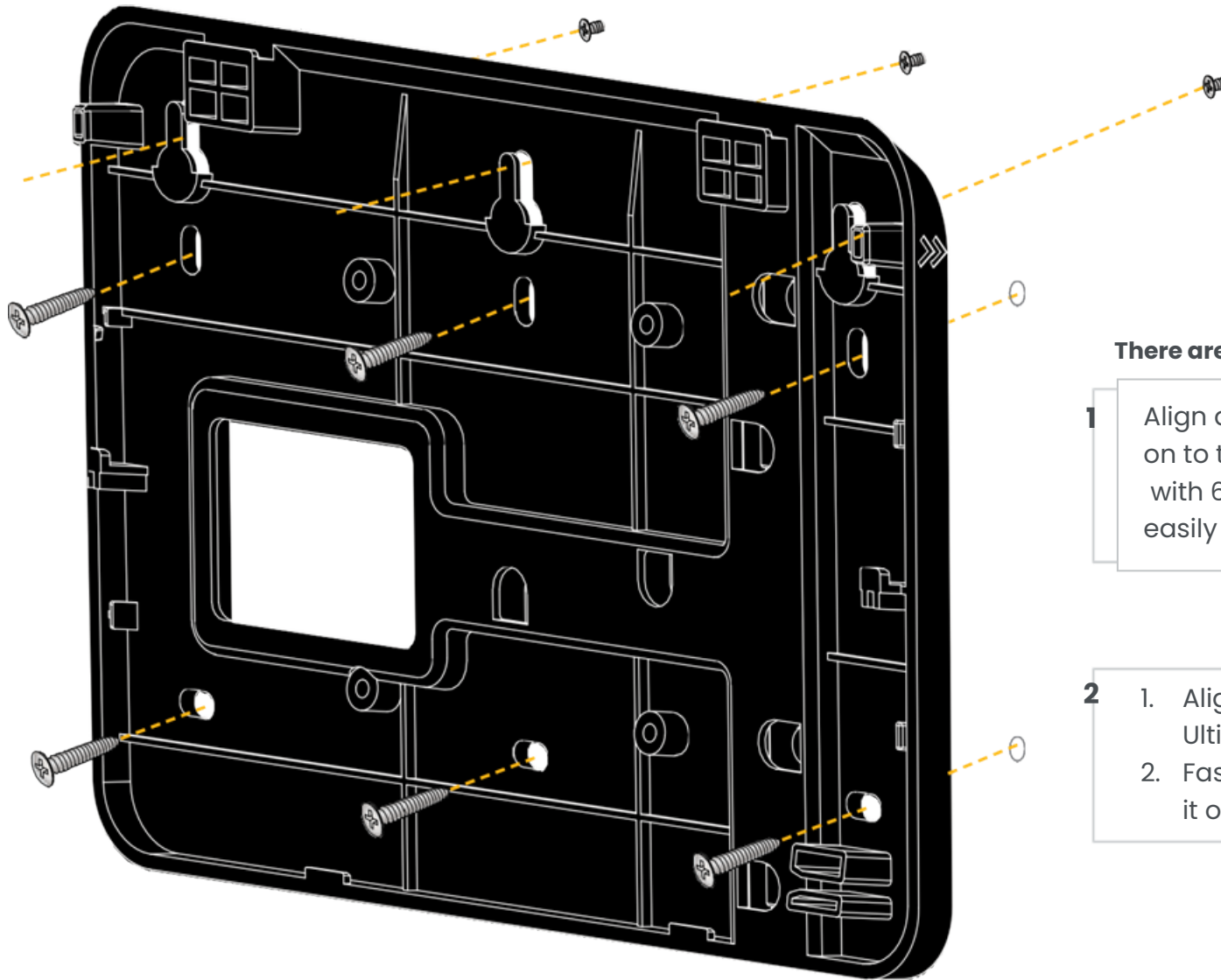
3/8 inch space

Opening Ultima 10 back cover (Mounting Plate)



1. Place Ultima 10 face down
2. Unlock the back cover (mounting plate) and gently open it

Caution Do not apply force



There are 2 ways to mount the back plate:

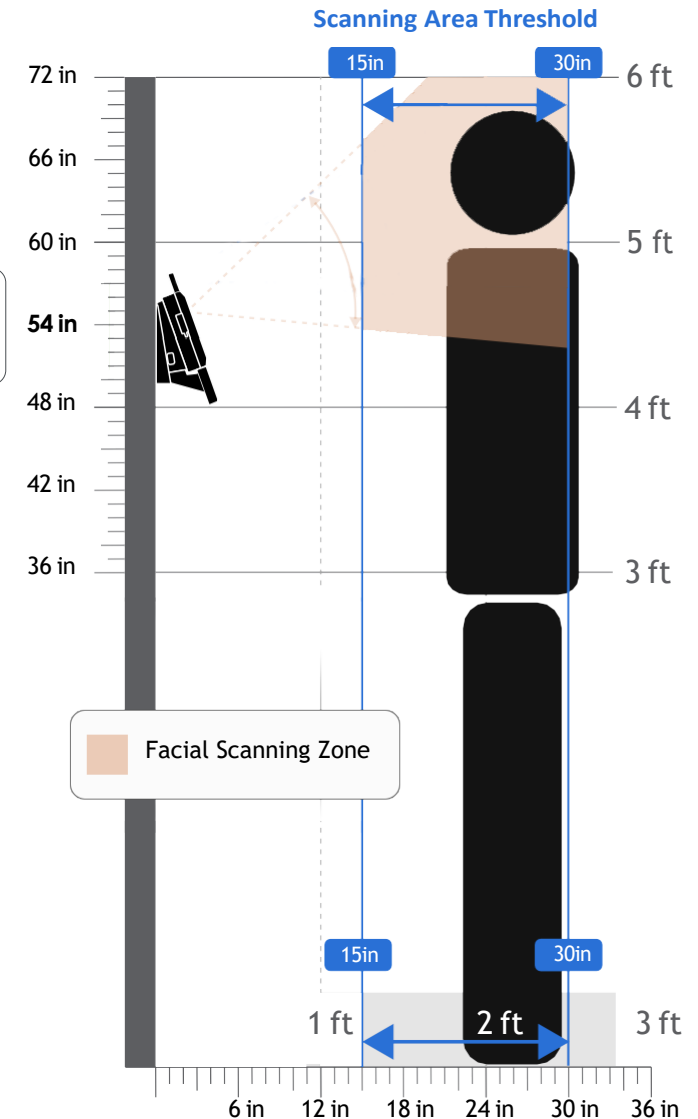
- 1 Align and hook up and hang it the Ultima 10 on to the wall. Without securing on the wall with 6 additional screws (This way you can easily remove it from the wall)
- 2
 1. Align and hook up and hang it the Ultima 10 on to the wall
 2. Fasten it with 6 bottom screws to secure it on the wall

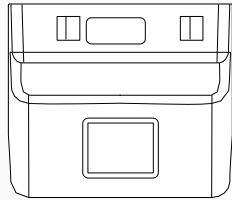


FOV (Field of view) for the combined facial sensors:

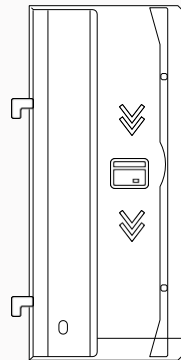
1. As per ADA regulations, ZKTeco WFM recommends installing the Ultima 10 clock between 54 inches from the top screw to 48 inches from the bottom above the ground
2. The “strike zone” in which the user should present their face for identification should be between 20 to 30 inches from the clock
3. The field of view for combined facial sensor fits anyone with a height of 4’08” to 6’00” while standing up straight between 20 to 30 inches in the “strike zone” from the clock
4. Anyone with a height of 6’00” to 6’06” should bend at their knees and place their face within the strike zone which is between 20 to 30 inches to fit inside the camera’s view angle for facial identification

Align the template top screw holes at 54 inch from the floor surface.

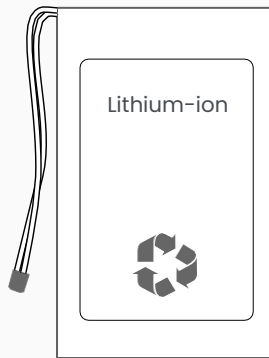




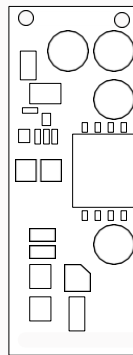
Fingerprint Reader



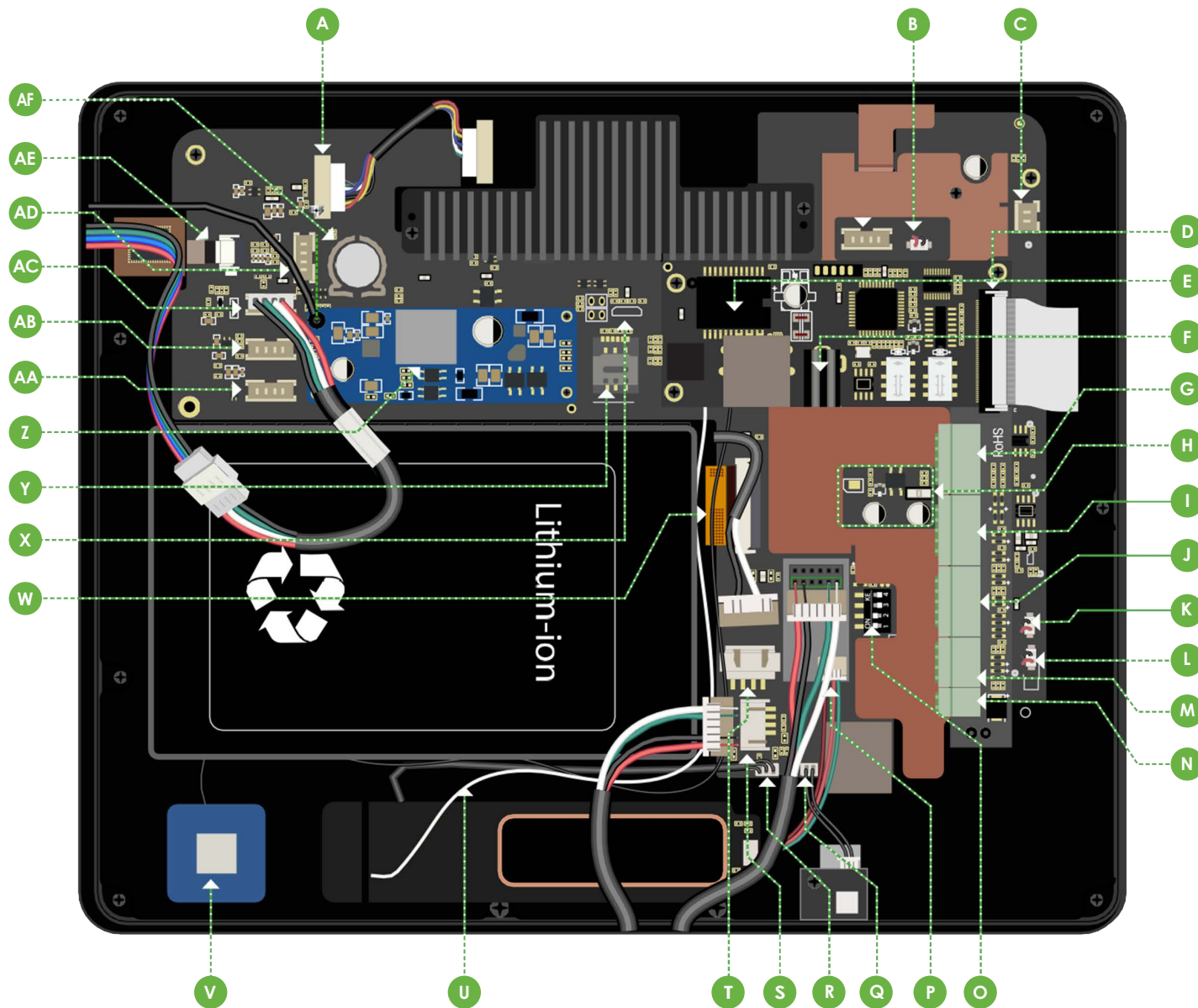
Card Reader



Battery Pack



PoE+



- A** Camera
- B** Main MIC Port
- C** 12V 2PIN Power IN
- D** Access Control and Power Board
- E** Ethernet Port
- F** 12V Power IN
- G** Relay Module
- H** 4G Antenna Port
- I** Wiegand IN 1
- J** Wiegand IN 2
- K** Secondary MIC Port
- L** Speaker
- M** Auxiliary Input
- N** 12V 200mA Power OUT
- O** RFID Selector Switch
- P** RFID Antenna
- Q** Reset
- R** Interactive LED
- S** Fingerprint
- T** Battery
- U** Wi-Fi Antenna
- V** GPS Antenna
- W** Display
- X** Debug miniUSB Port
- Y** SIM Card
- Z** PoE+ Module
- AA** Reserved USB Port
- AB** Magnetic Reader
- AC** Barcode Reader
- AD** Reserved UART Port
- AE** Touch Screen
- AF** RTC Button Battery

Relay module connectors

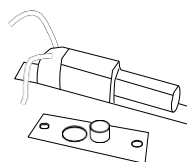
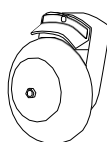


Warning Relay is rated for **12V only** and **should not exceed the recommended voltage**. If connecting a higher voltage device some extra components maybe necessary. Please consult ZKTeco WFM for further guidance before connecting to higher voltages.



Connecting Wires

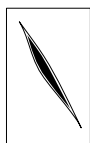
Please follow components manufacturer installation instruction



NC
COM
NO



NC
COM
NO



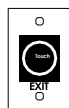
**12 Volt
Components**

12V
GND
IWD 0
IWD 1
BUZ
GLED

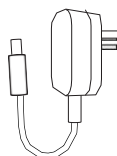


**5 Volt
Components**

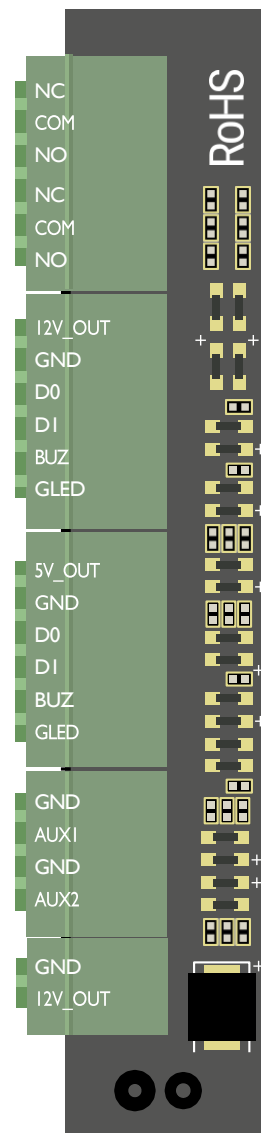
5V
GND
IWD 0
IWD 1
BUZ
GLED

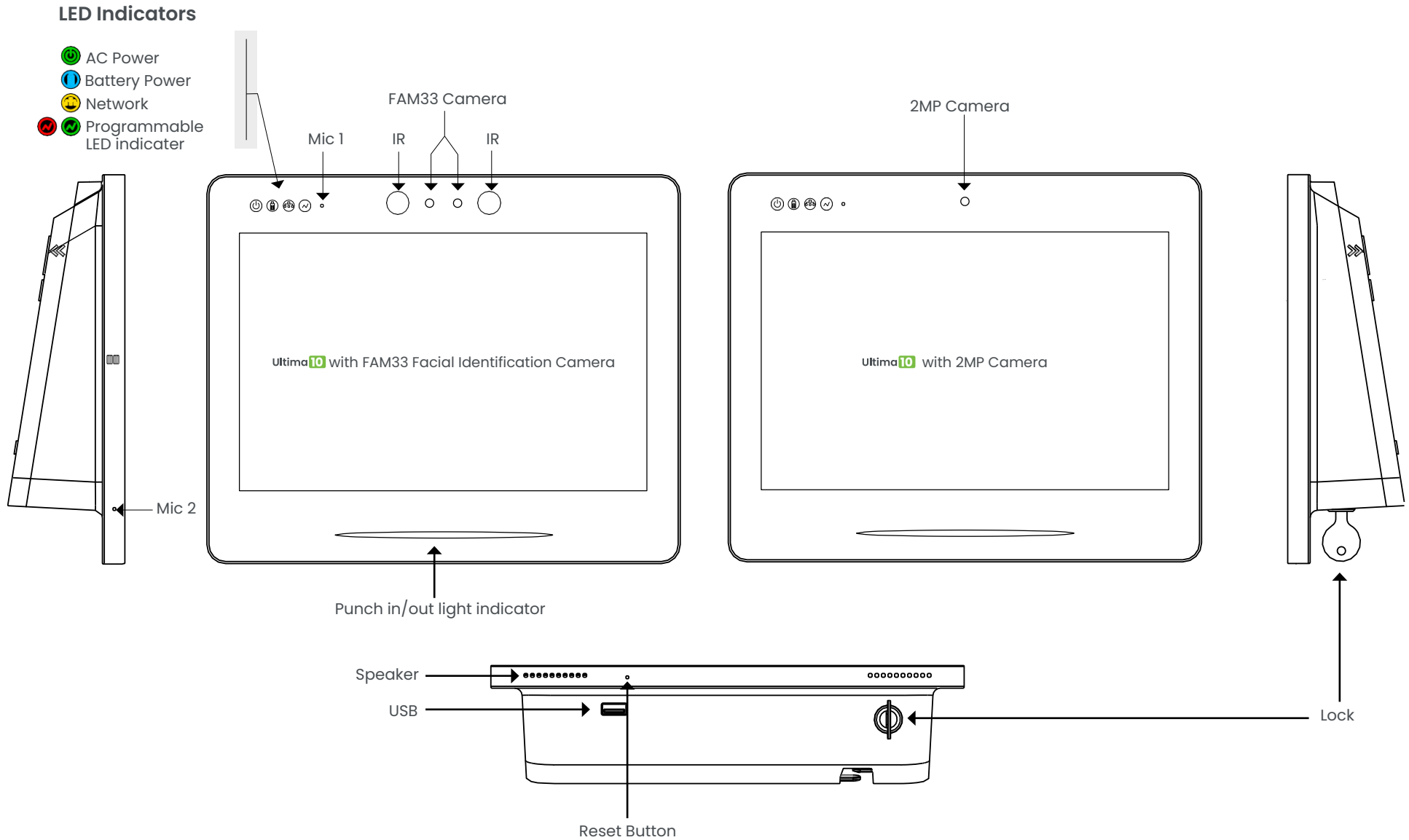


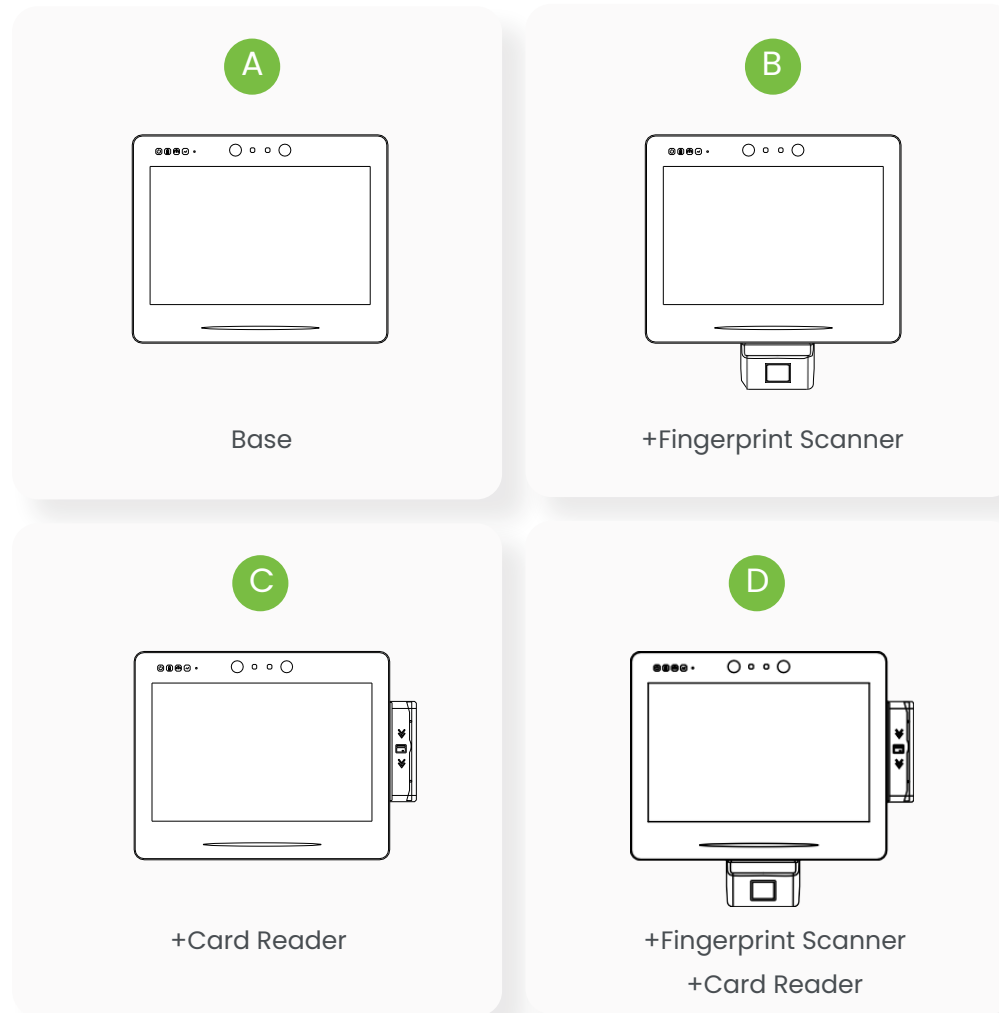
GND
AUX 1
GND
AUX 2



GND
12V DC







Warning The Ultima 10 can support multiple readers at a time i.e. MagStripe or Barcode or RFID module, but not combinations of the two. Please note: The Magstripe and Barcode module is always installed/configured to be at the far end of the Ultima terminal.

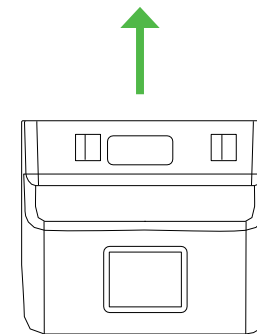
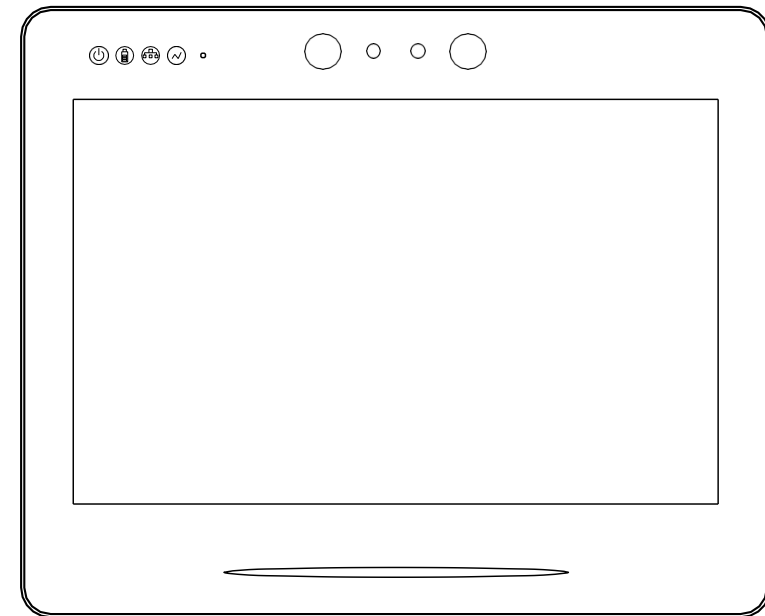
Step 1: Opening the back cover

Step 2: Removing screws

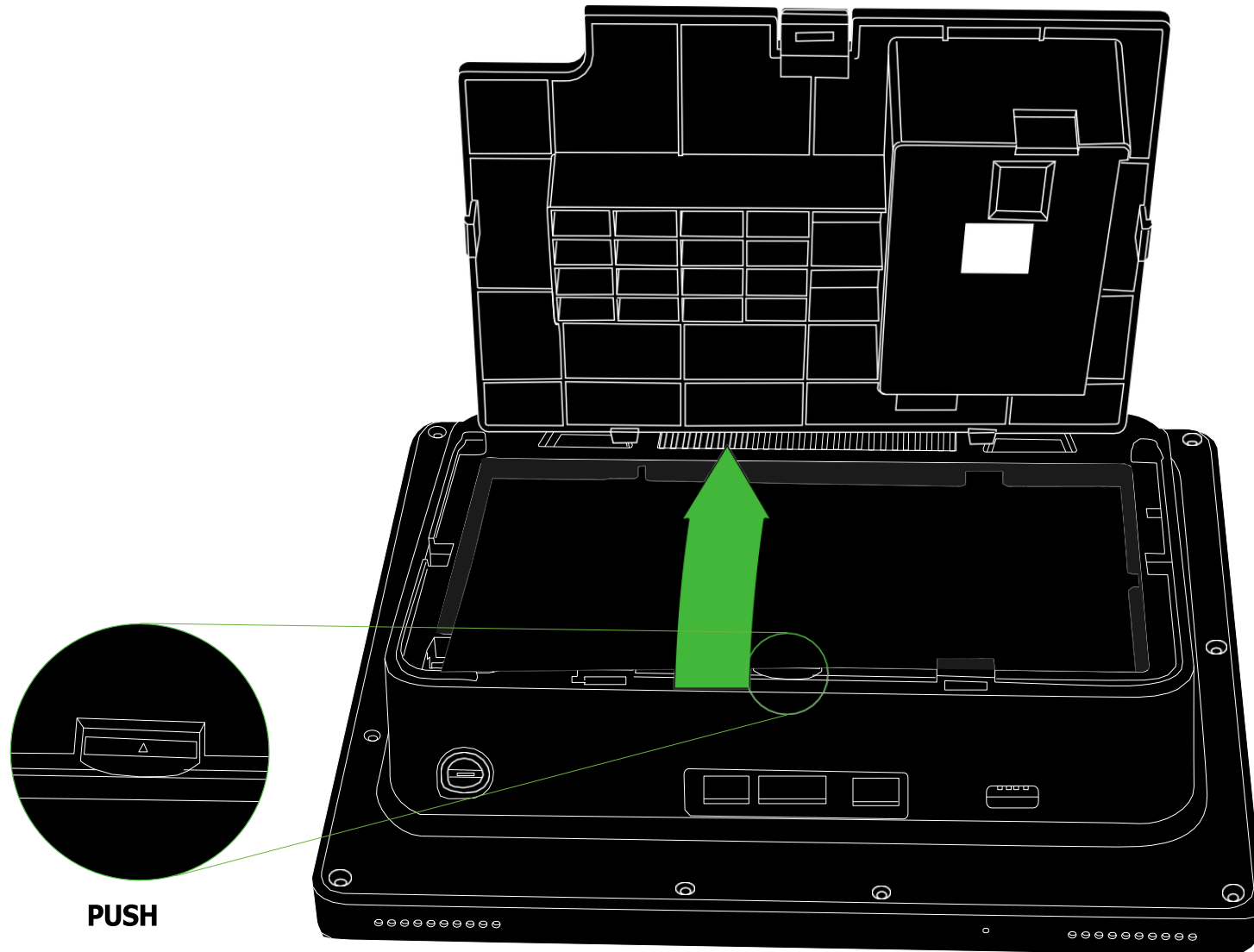
Step 3: Inserting wires and assembling

Step 4: Connecting wires to motherboard

Step 5: Testing



F - Fingerprint scanner **R** - RFID module



Push the latch lock and lift the back cover upward to open assembly completely to access to internal component.

Caution Do not apply force

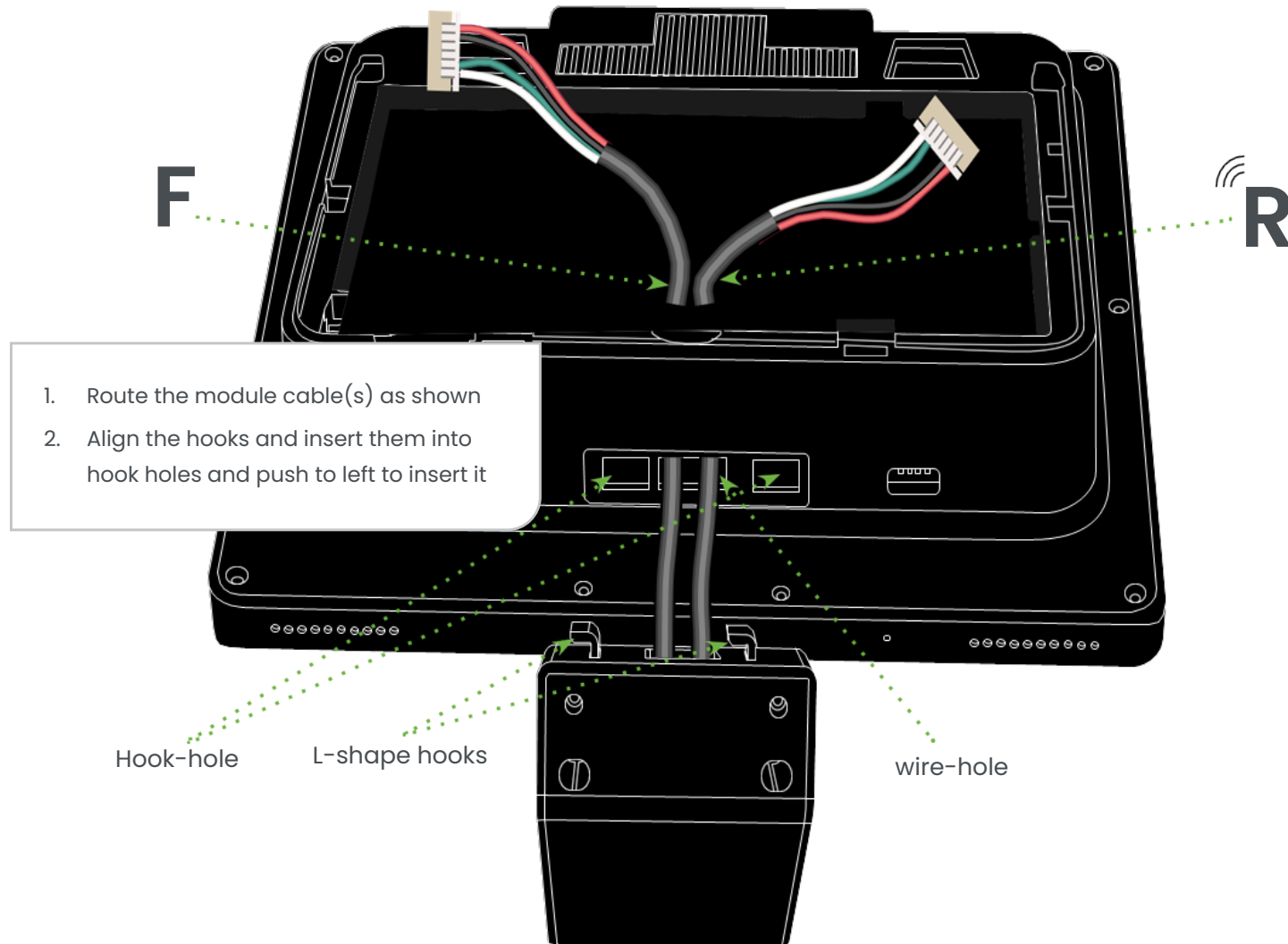
Installing Fingerprint scanner with RFID Module |

Removing screws

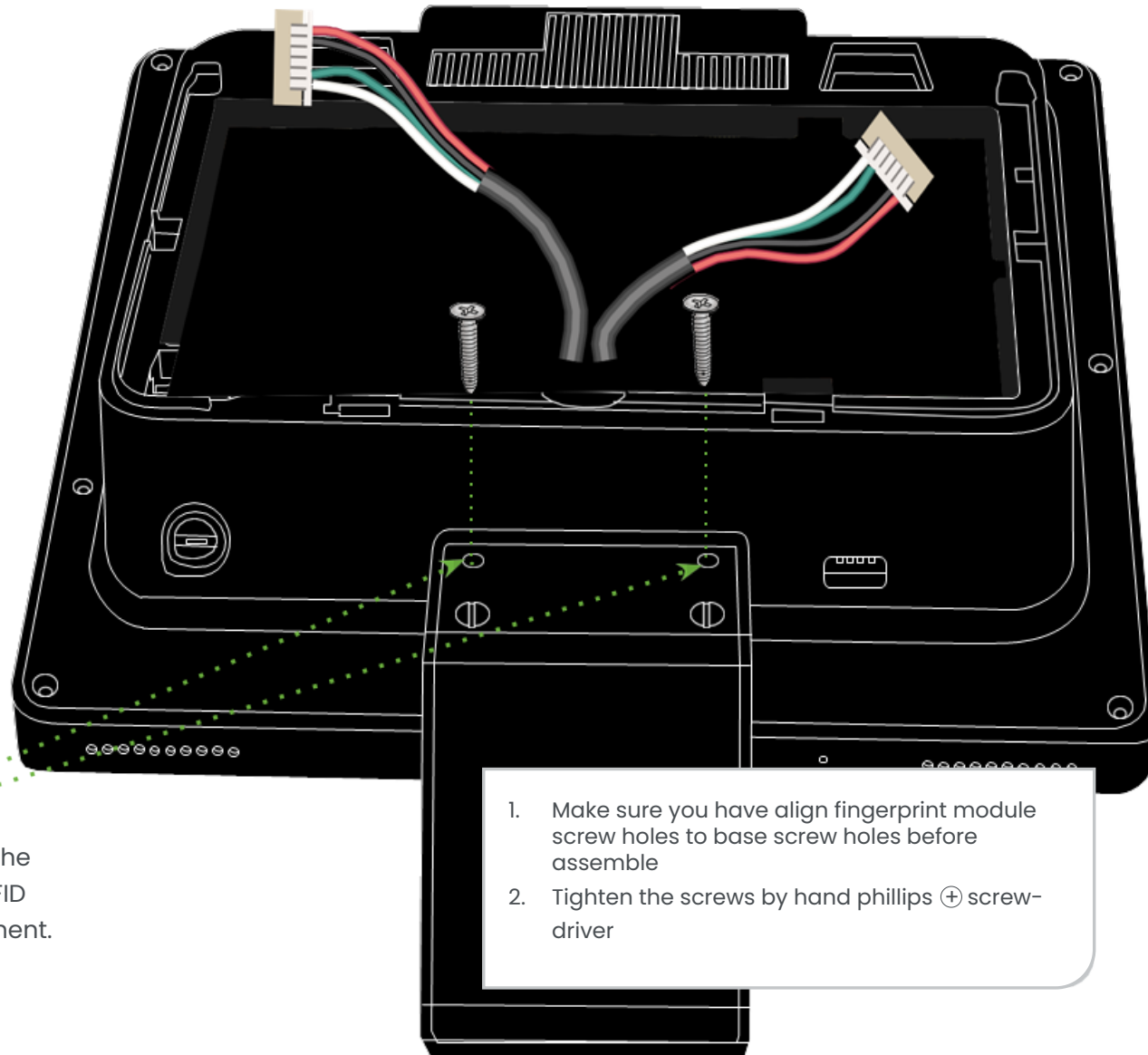


Un-screw and remove 2 screws

Installing Fingerprint scanner with RFID Module | Inserting wires and assembling



Installing Fingerprint scanner with RFID Module | Inserting wires and assembling

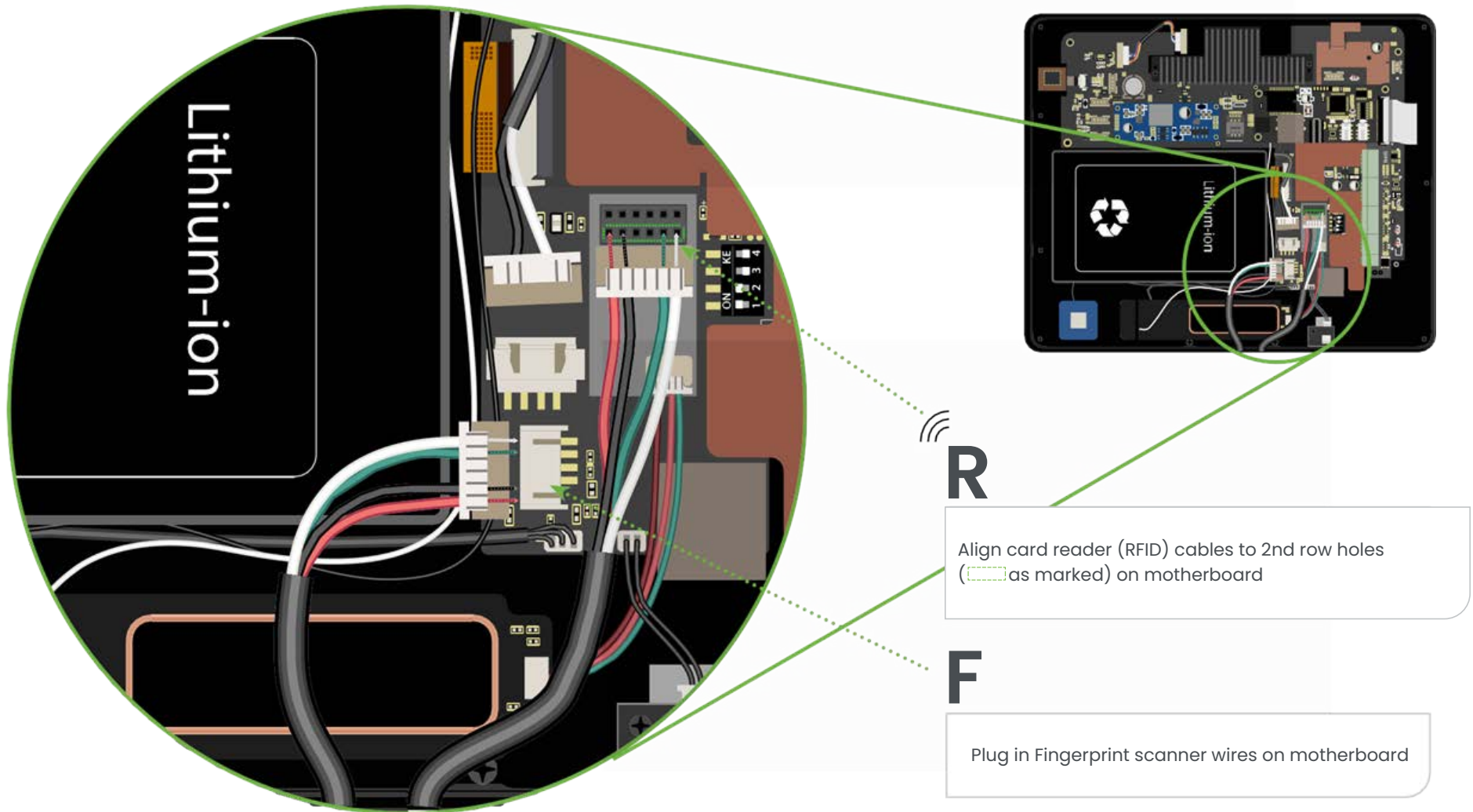


Screw holes for mounting the fingerprint scanner with RFID module to the base component.

1. Make sure you have align fingerprint module screw holes to base screw holes before assemble
2. Tighten the screws by hand phillips $\oplus$ screw-driver

Installing Fingerprint scanner with RFID Module |

Connecting wires to motherboard



Installing Fingerprint scanner with RFID Module |

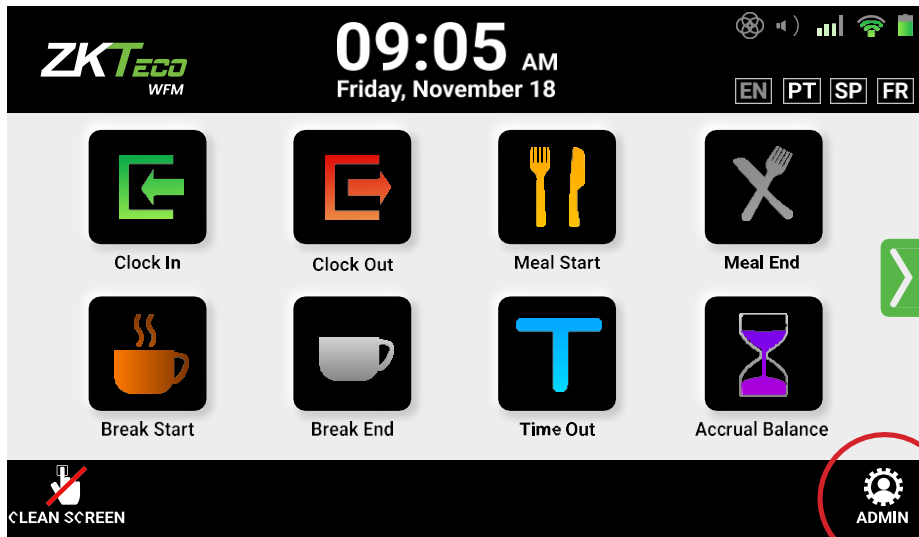
Closing back cover



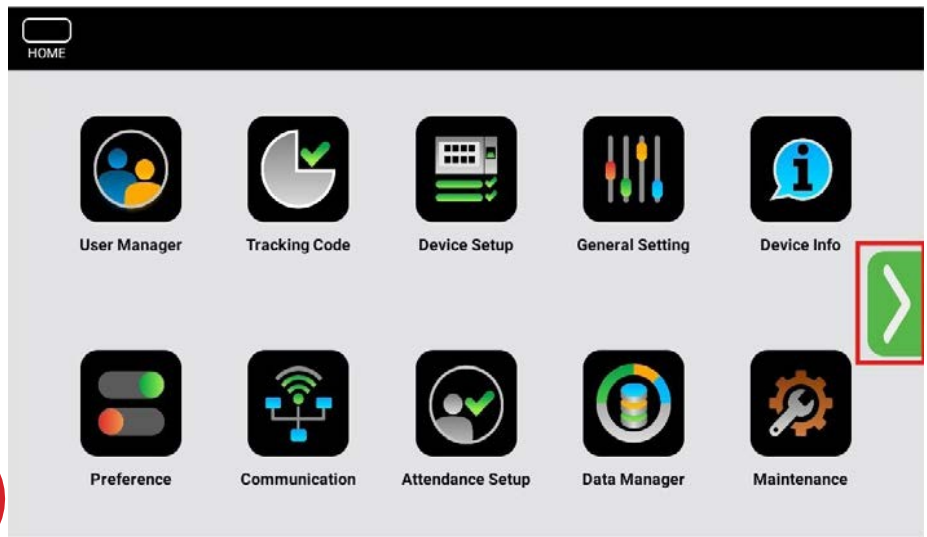
Close the back cover

Installing Fingerprint scanner with RFID Module |

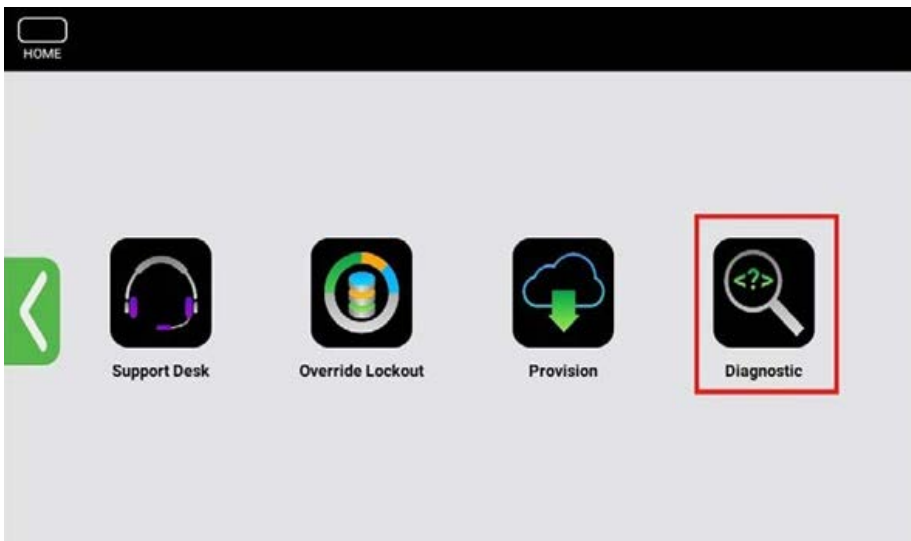
Testing fingerprint module



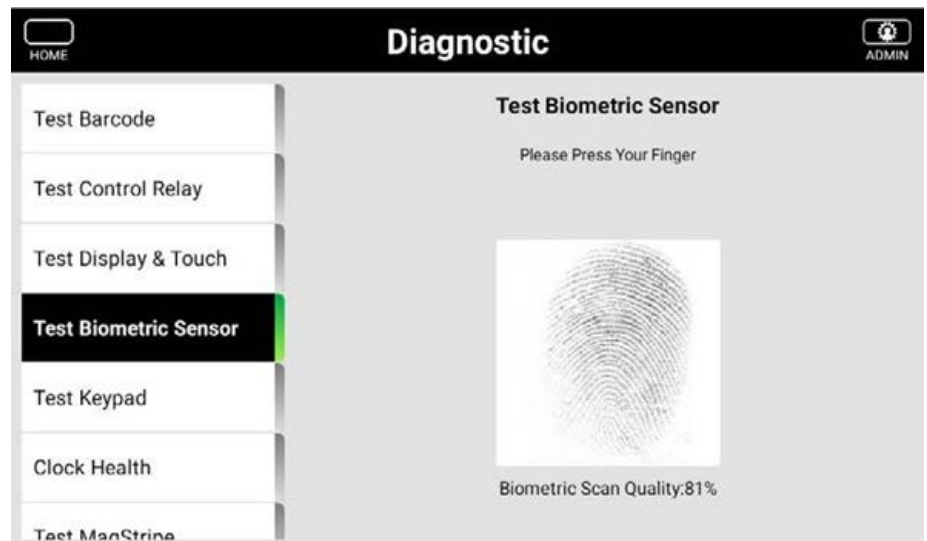
Press Admin  to login



Press next screen 



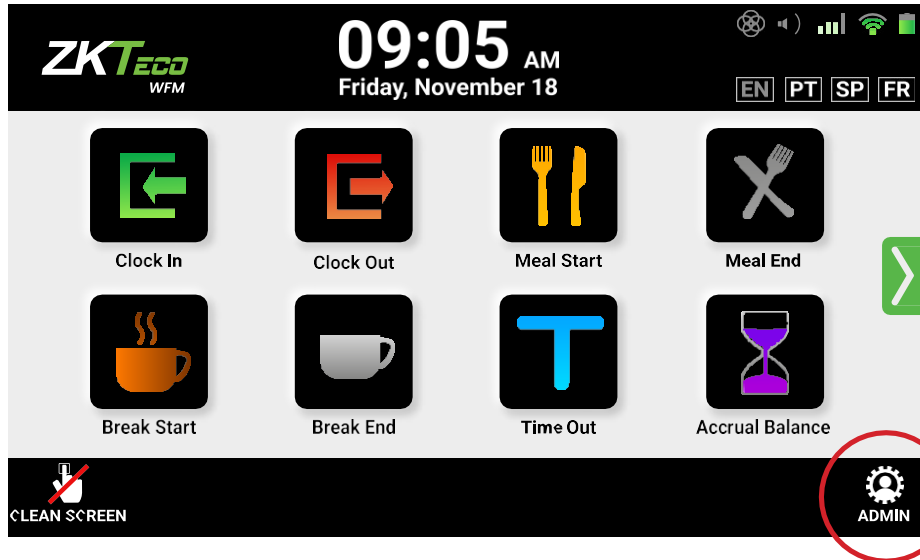
Press Diagnostic



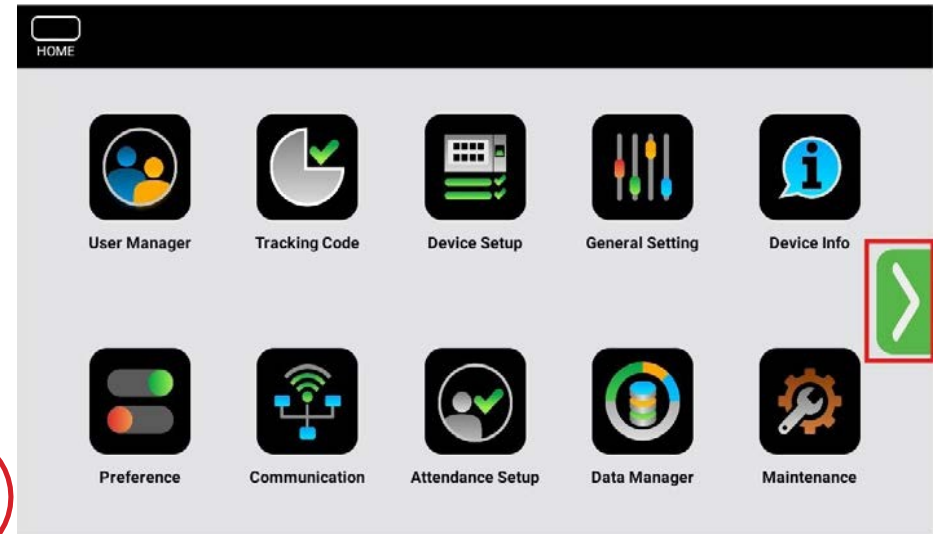
Test Biometric Sensor

Installing Fingerprint scanner with RFID Module |

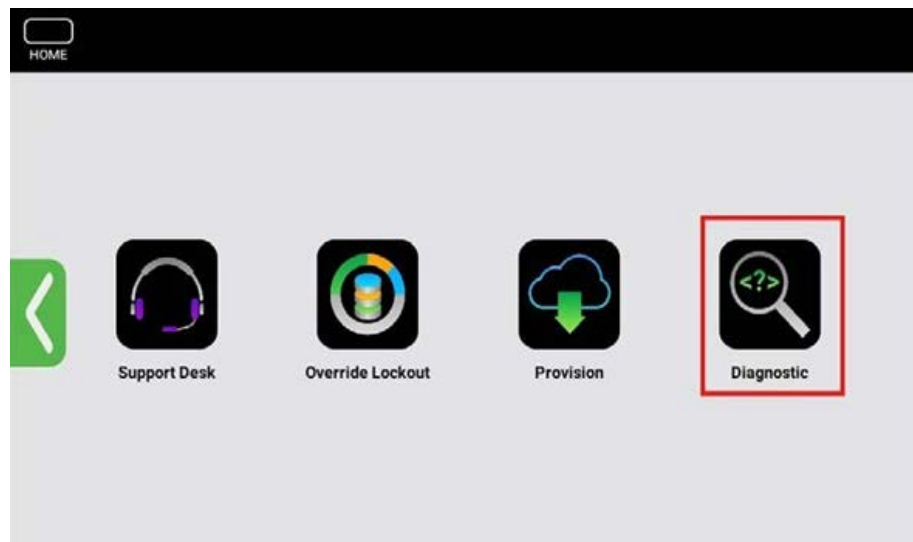
Testing RFID module



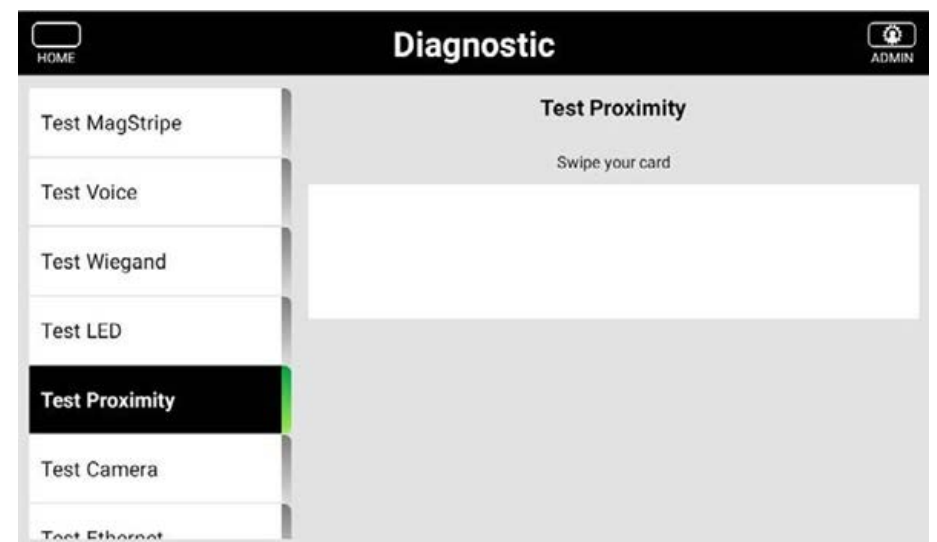
Press Admin  to login



Press next screen 



Press Diagnostic



Test Proximity

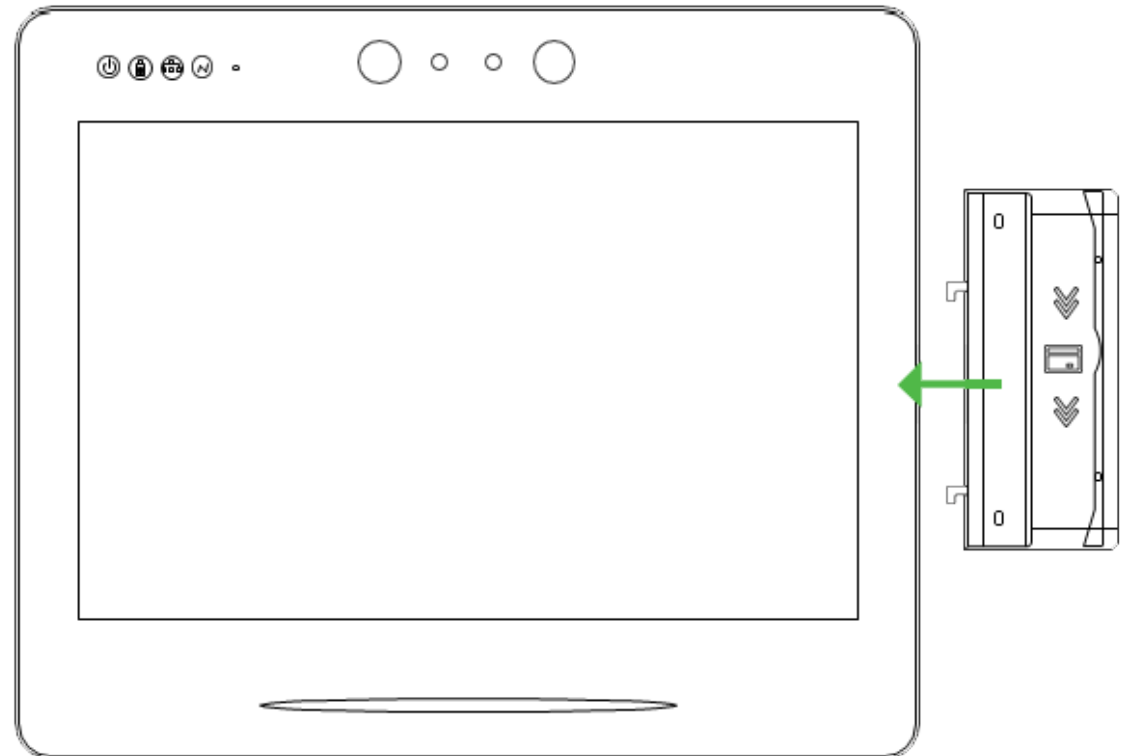
Step 1: Opening the back cover

Step 2: Removing screws

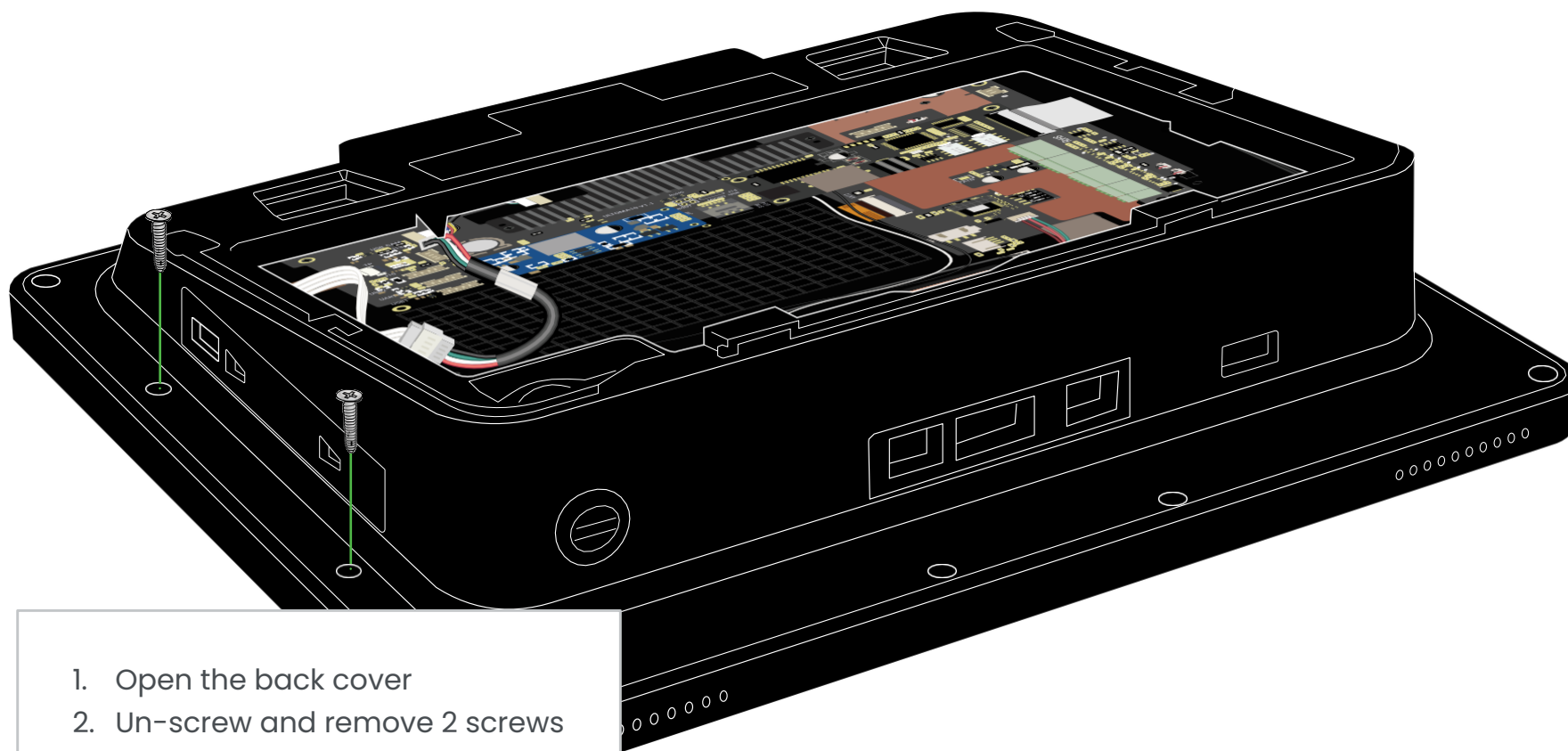
Step 3: Inserting wires and assembling

Step 4: Connecting wires to motherboard

Step 5: Testing

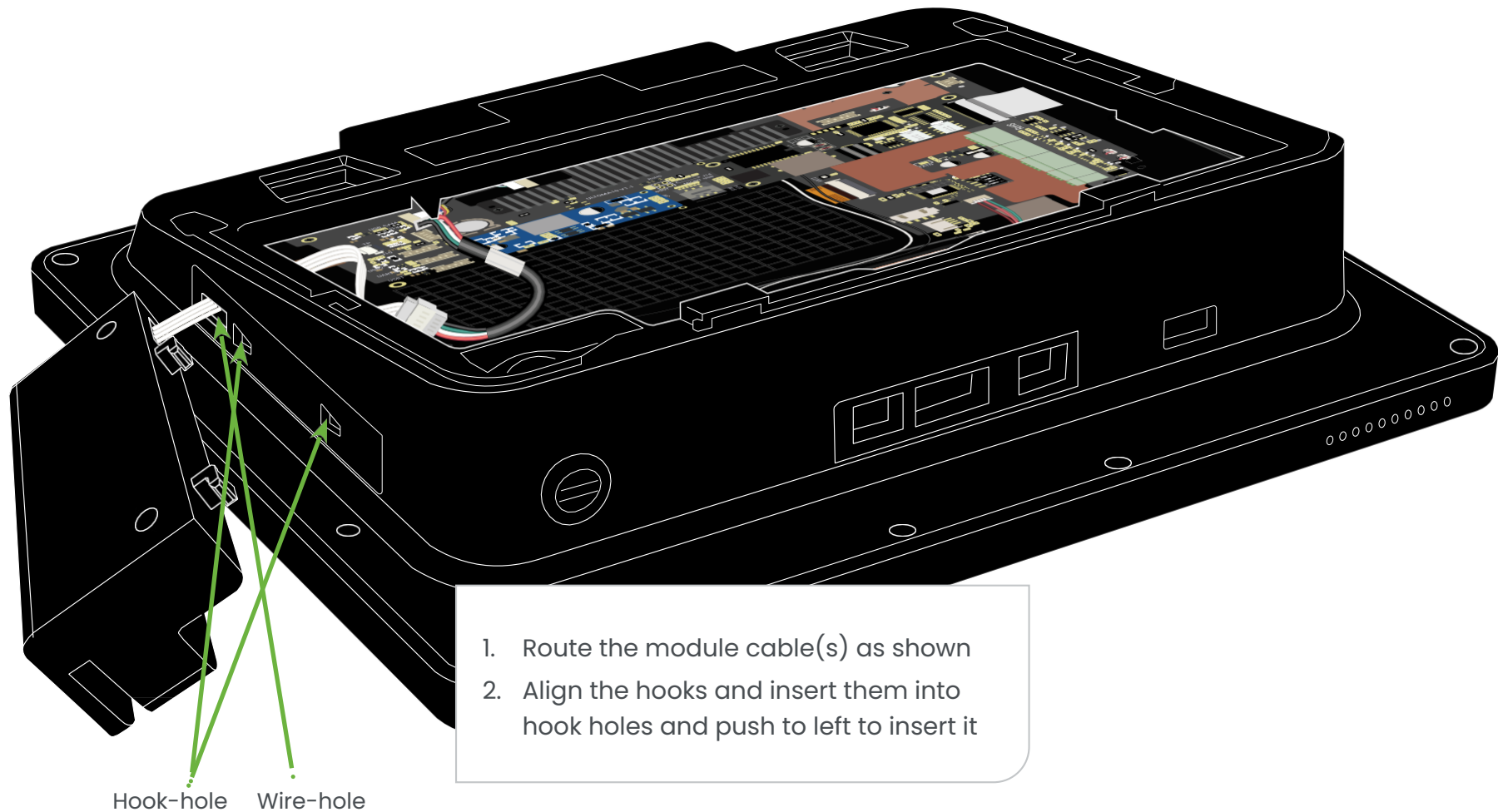


M - Magnetic card reader module

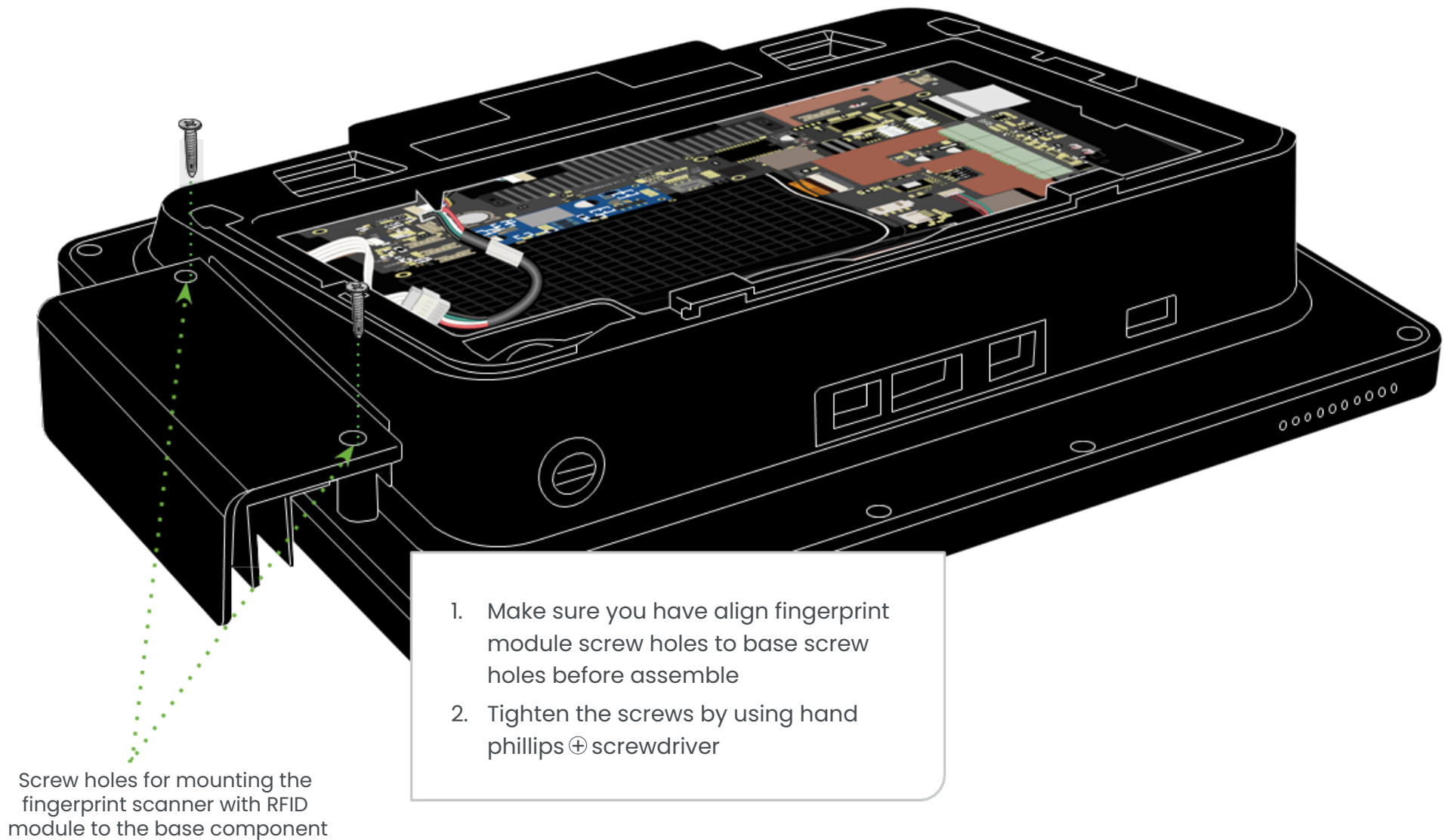


1. Open the back cover
2. Un-screw and remove 2 screws

Installing Magnetic card reader Module | Inserting wires and assembling

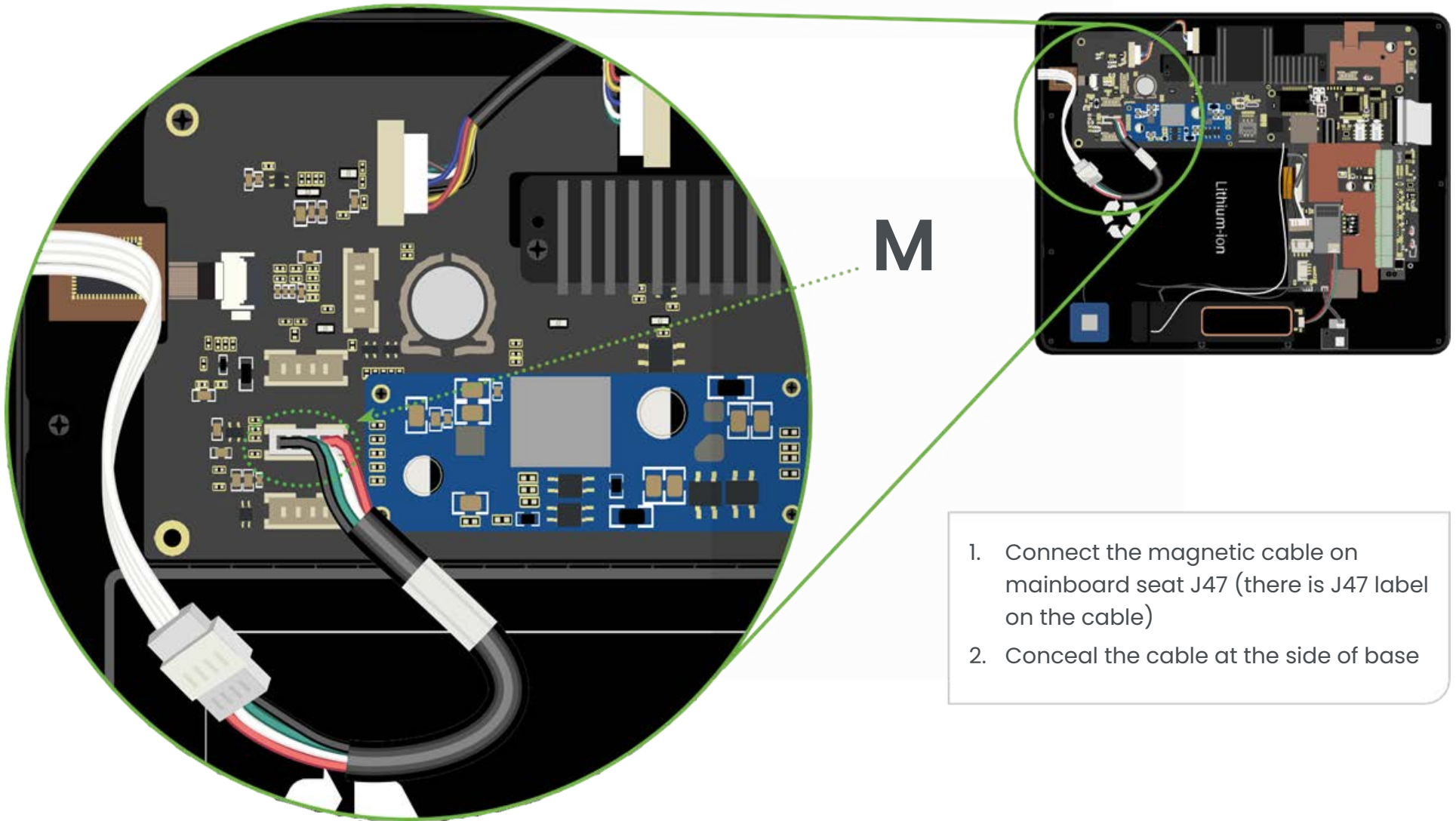


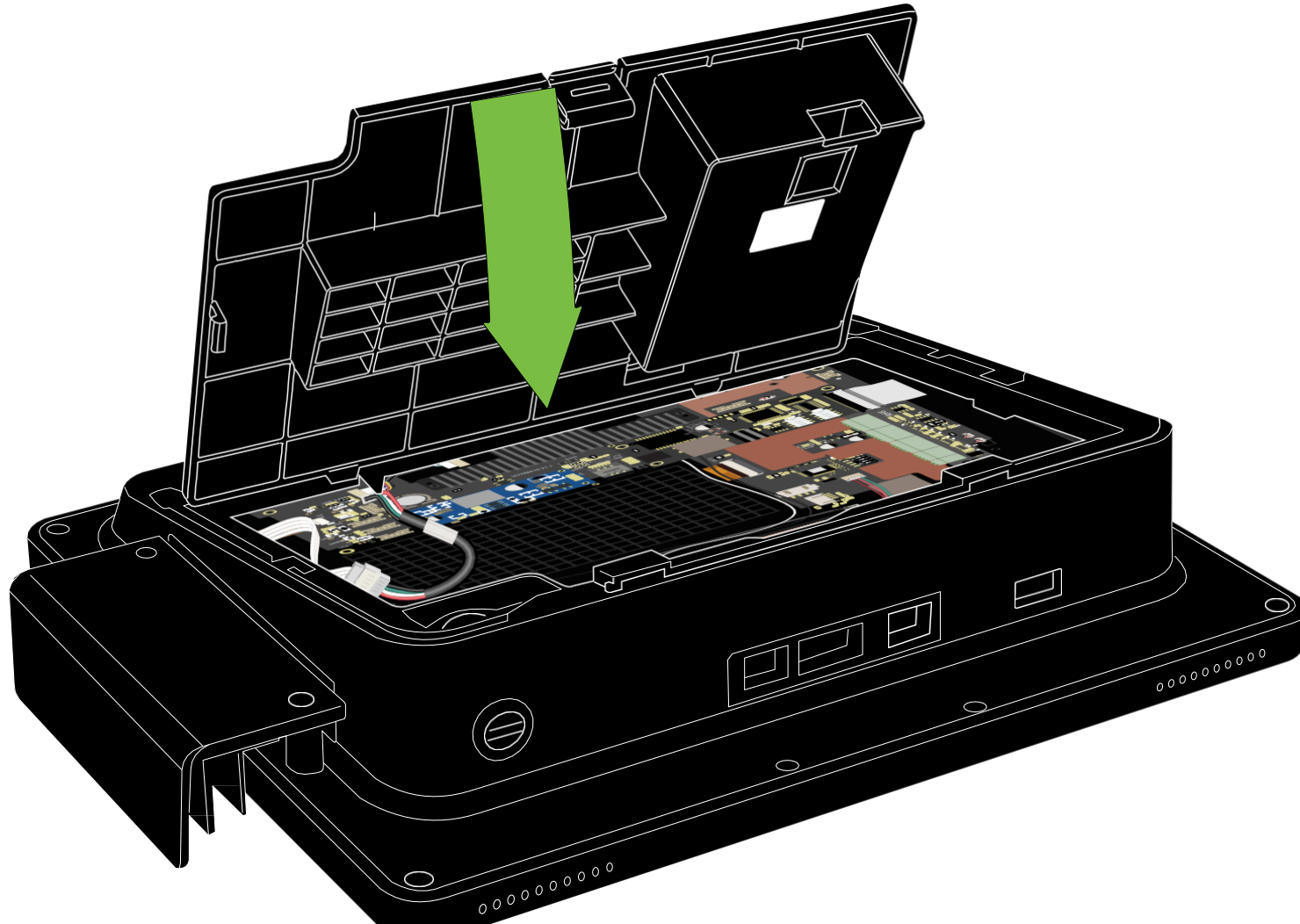
Installing Magnetic card reader Module | Inserting wires and assembling



Installing Magnetic card reader Module | Inserting wires and assembling

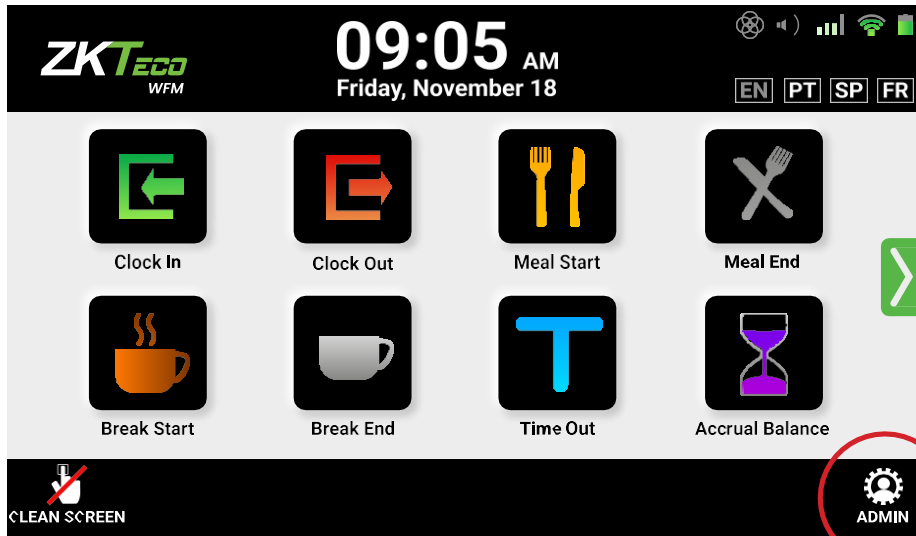
M



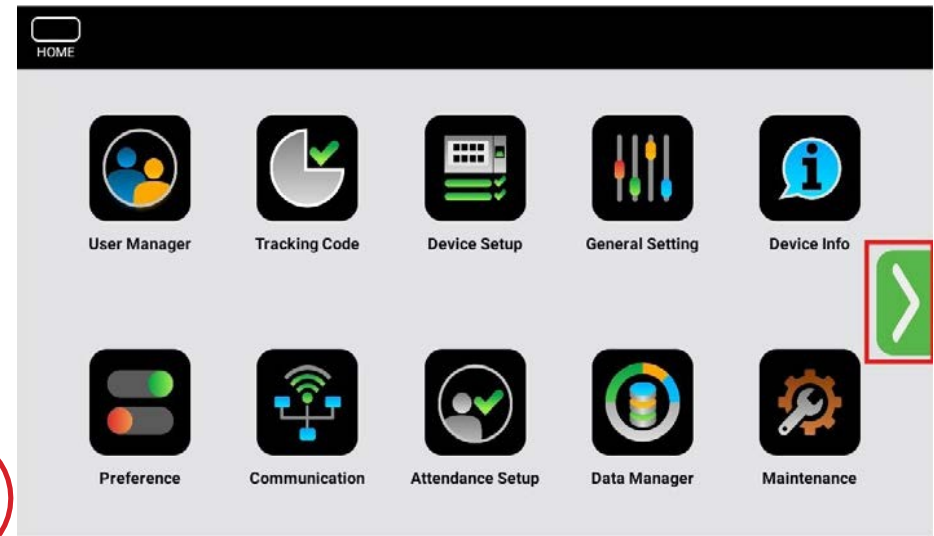


Close the back cover

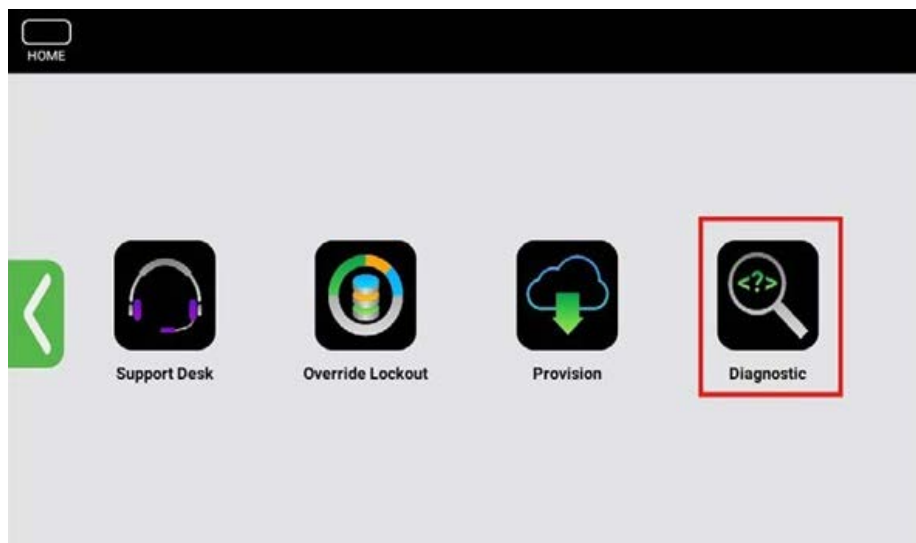
Installing Magnetic card reader Module | Testing Magnetic module



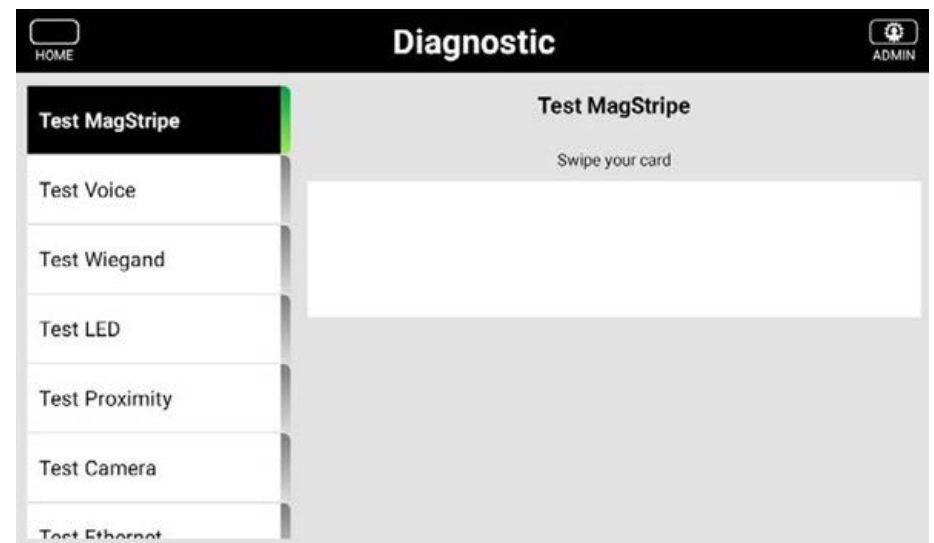
Press Admin  to login



Press next screen >



Press Diagnostic



Test MagStripe

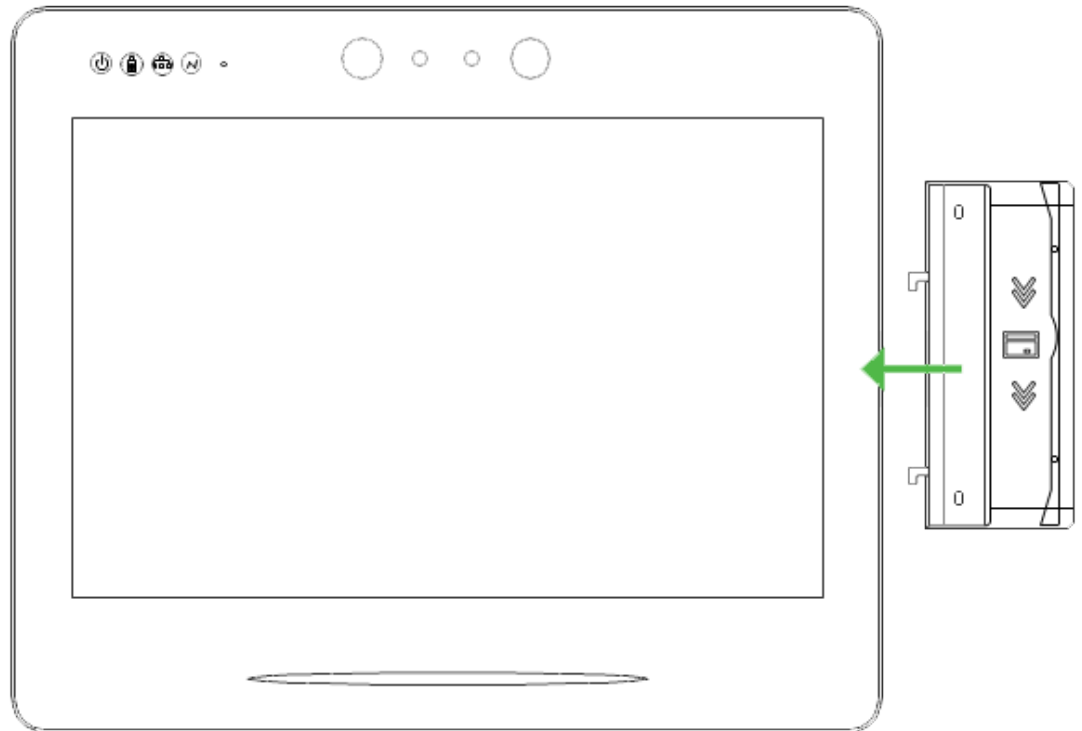
Step 1: Opening the back cover

Step 2: Removing screws

Step 3: Inserting wires and assembling

Step 4: Connecting wires to motherboard

Step 5: Testing



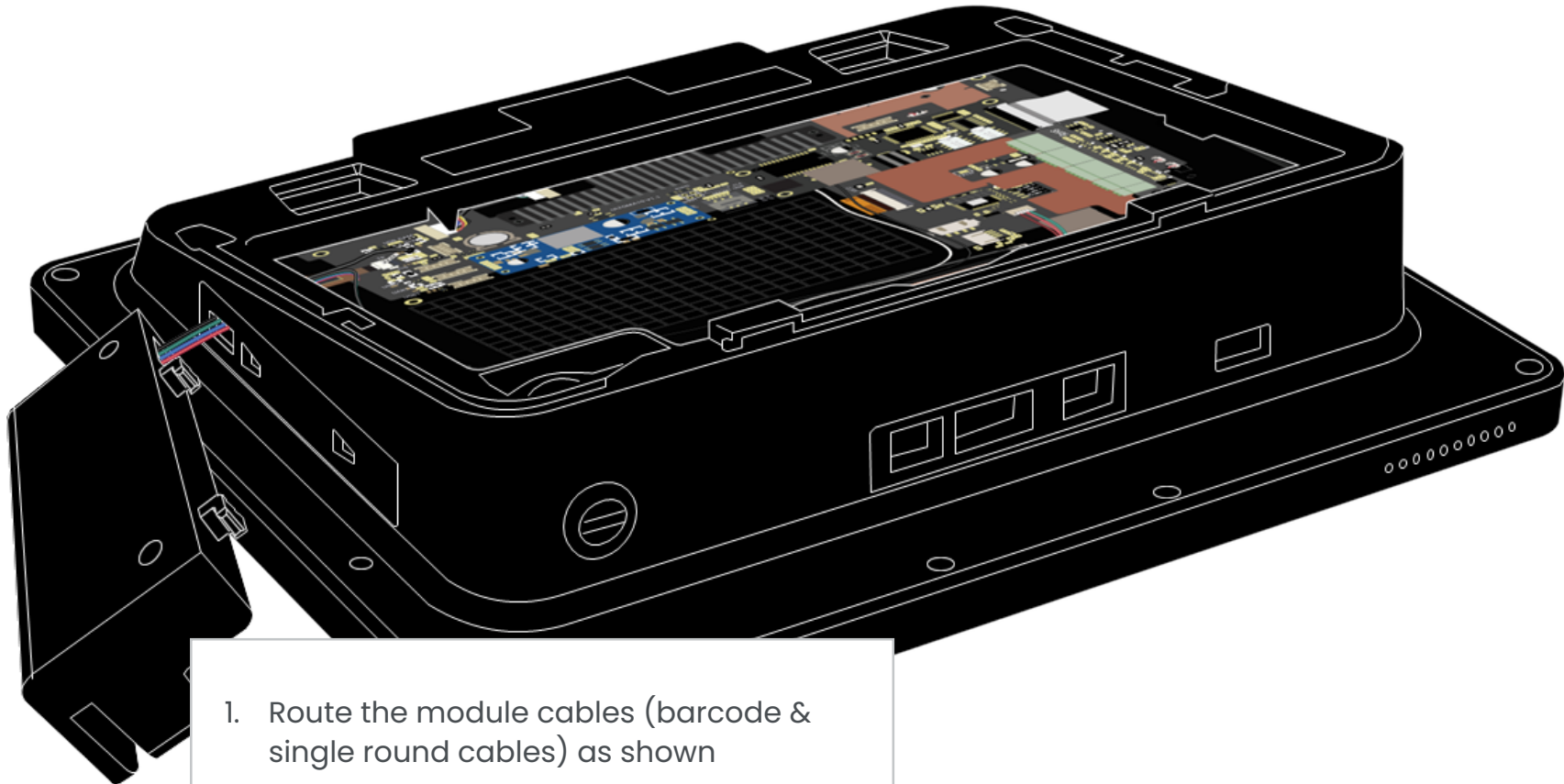
B – Barcode Card Reader Module



1. Open the back cover
2. Un-screw and remove 2 screws

Installing Barcode card reader Module | Inserting wires and assembling

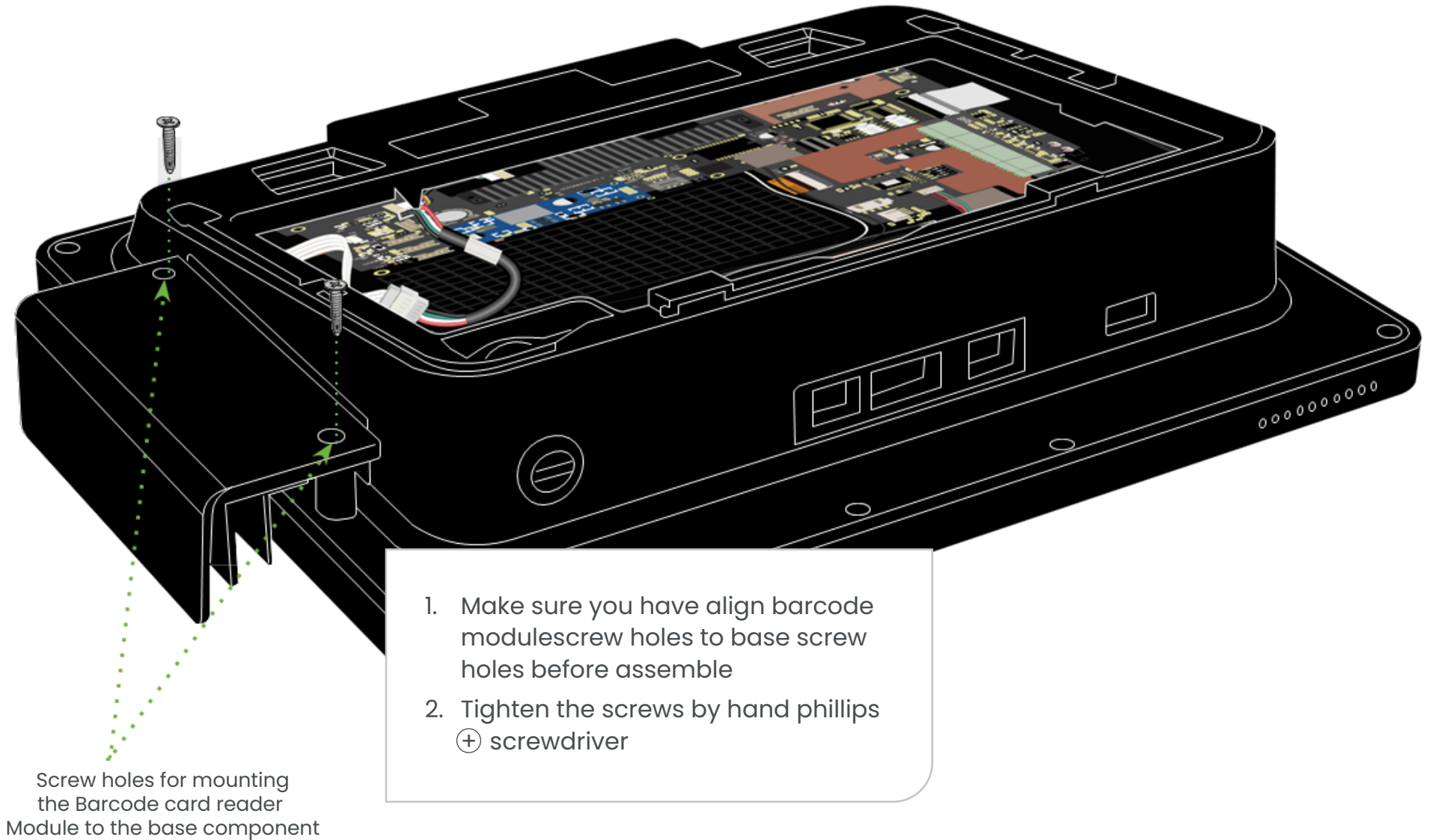
B



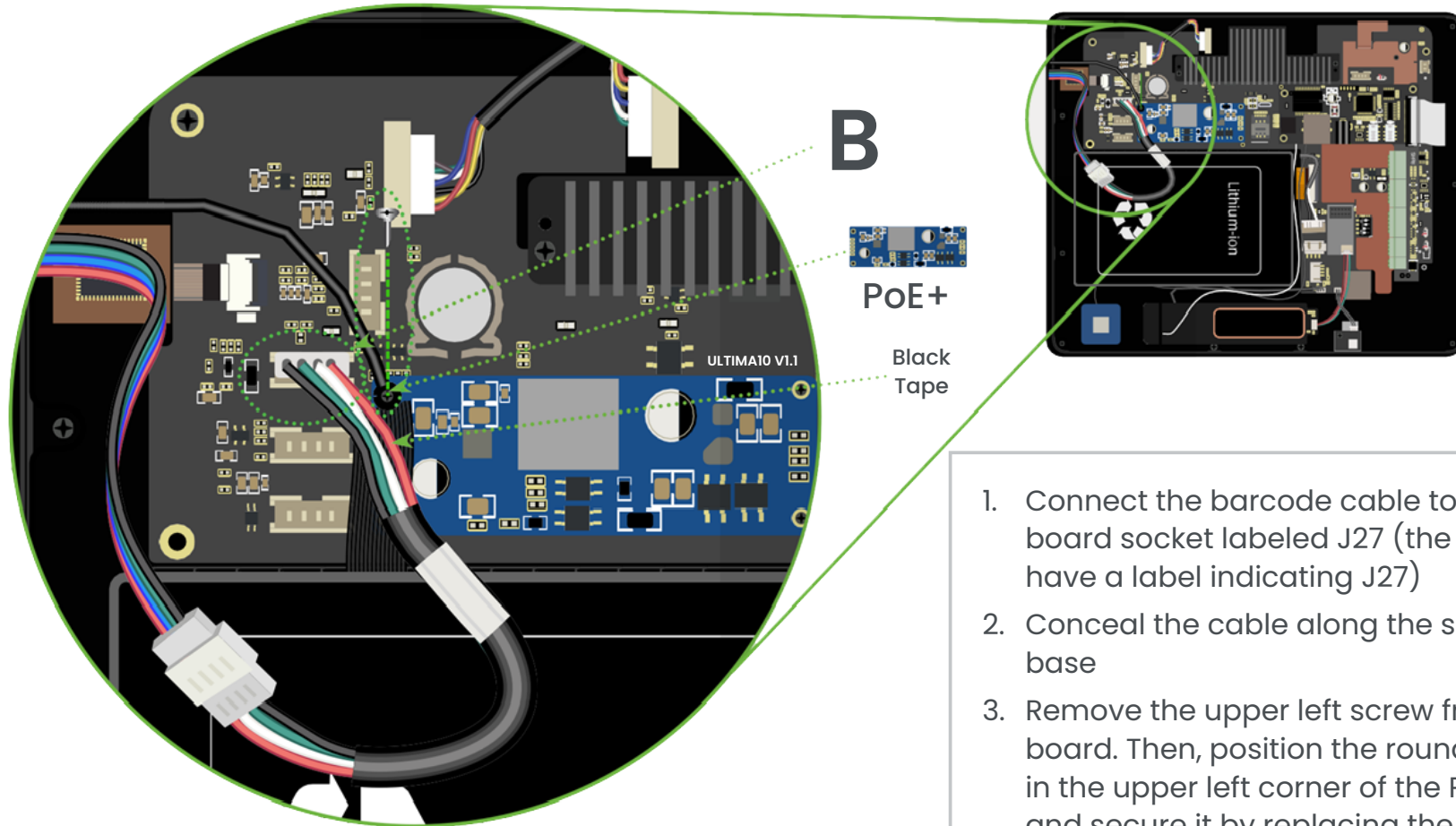
1. Route the module cables (barcode & single round cables) as shown
2. Align the hooks and insert them into hook holes and push to down to insert it

Installing Barcode card reader Module | Inserting wires and assembling

B



Installing Barcode card reader Module | Inserting wires and assembling

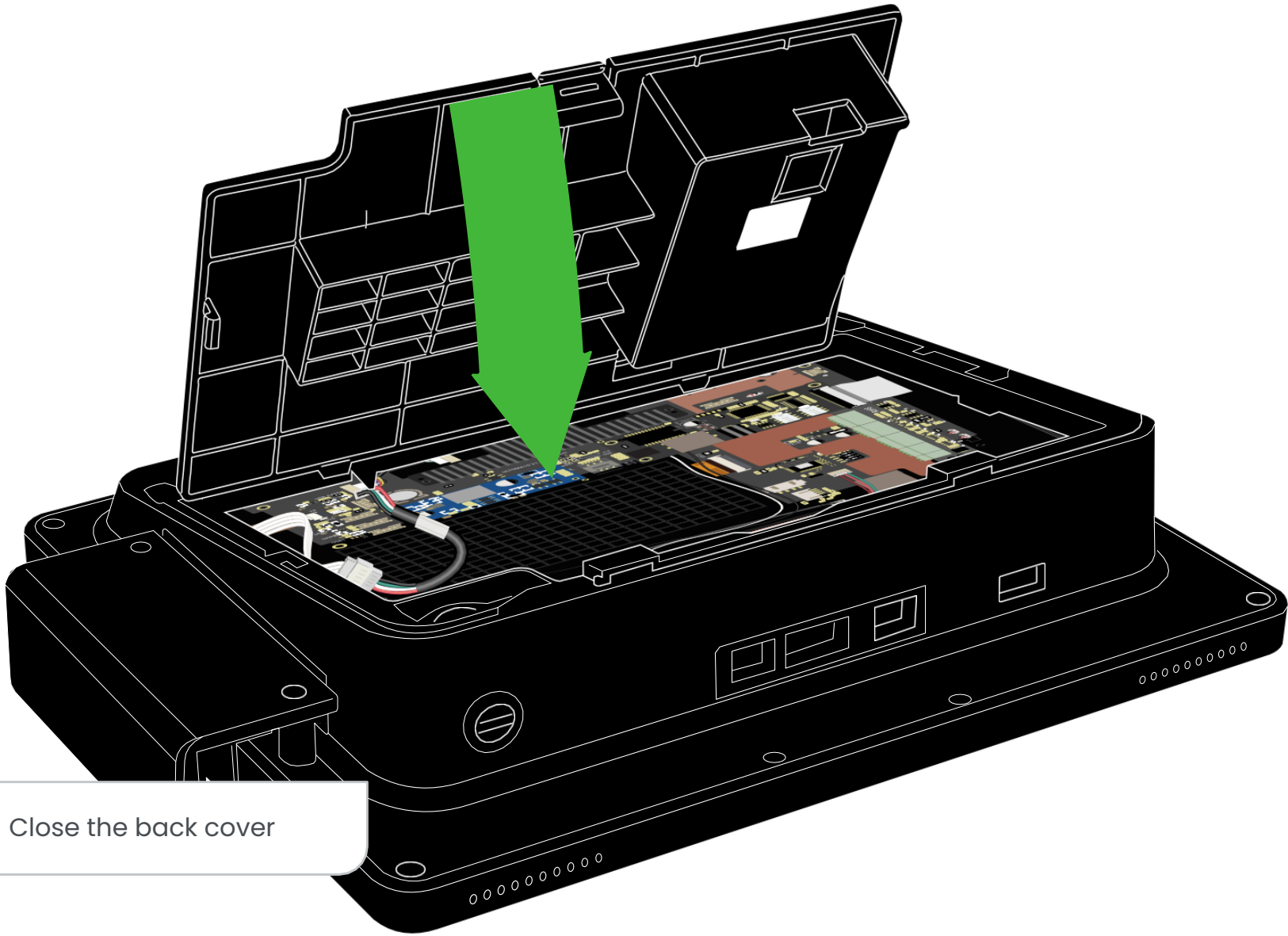


1. Connect the barcode cable to the main-board socket labeled J27 (the cable will have a label indicating J27)
2. Conceal the cable along the side of the base
3. Remove the upper left screw from the PoE+ board. Then, position the round single cable in the upper left corner of the PoE+ board and secure it by replacing the screw (as shown)
4. Cover the left side of the PoE+ board with the supplied **black tape**

Installing Barcode card reader Module |

Closing back cover

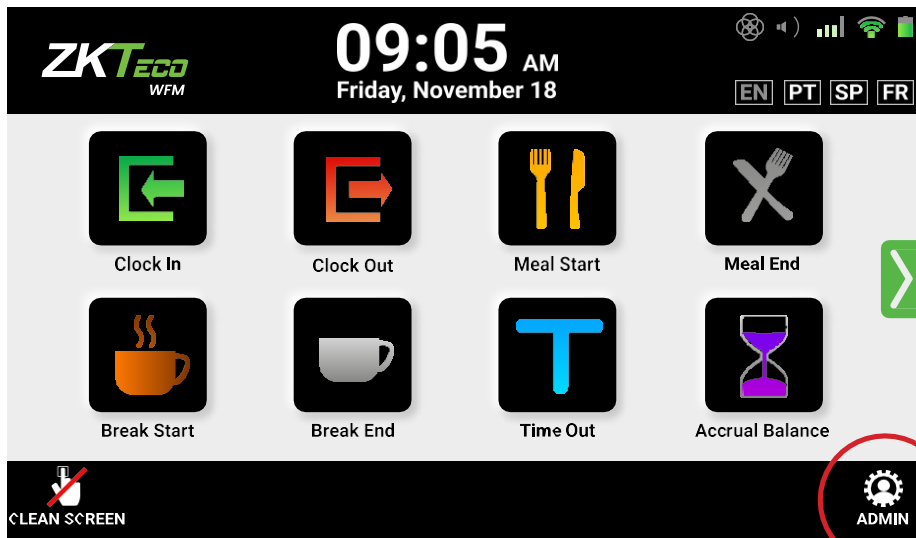
B



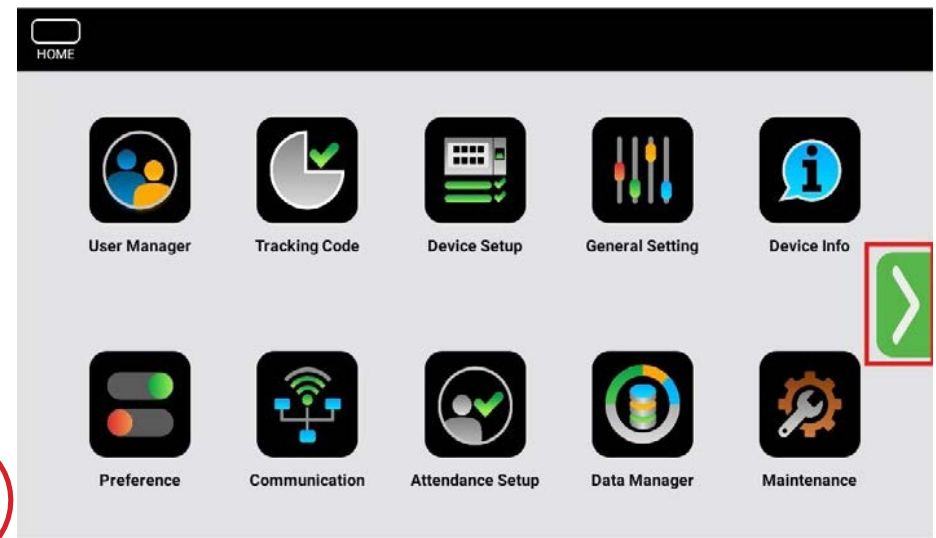
Close the back cover

Installing Barcode card reader Module | Testing Barcode Module

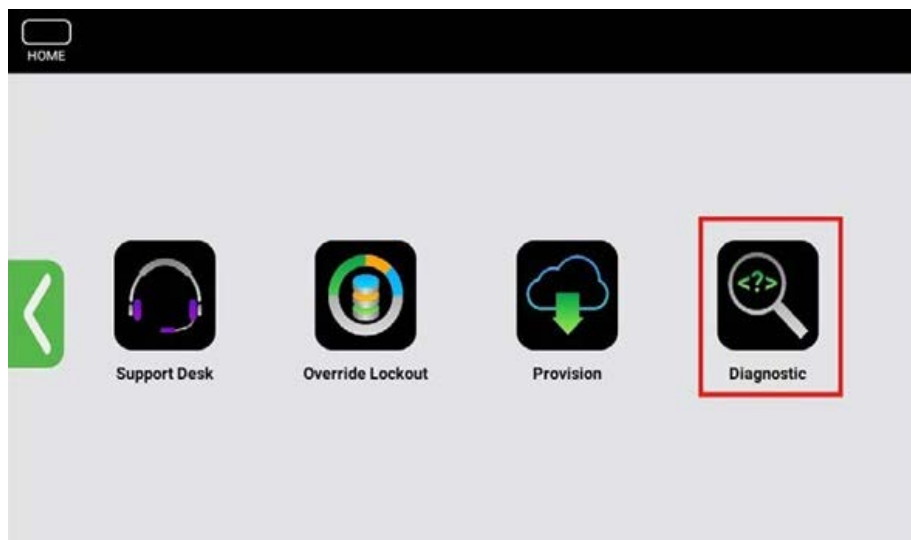
B



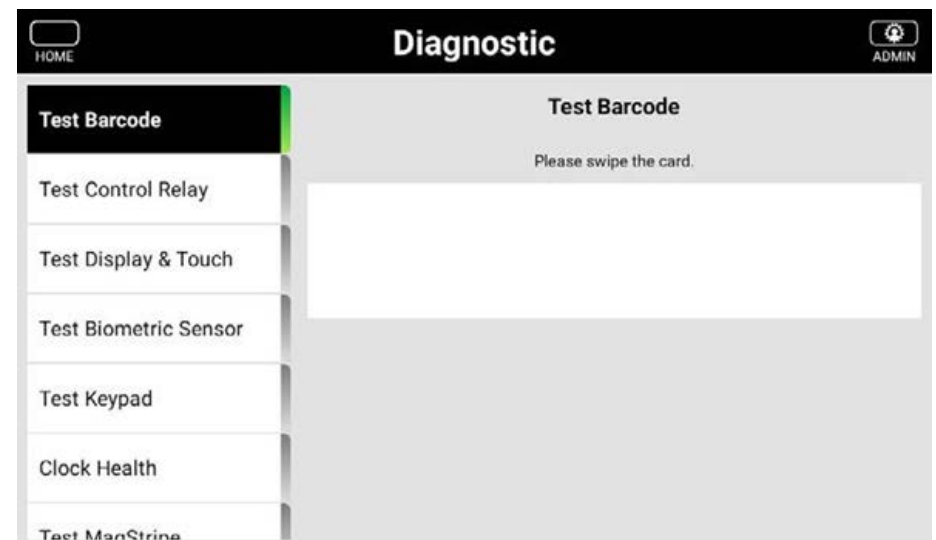
Press Admin  to login



Press next screen 



Press Diagnostic



Test Barcode

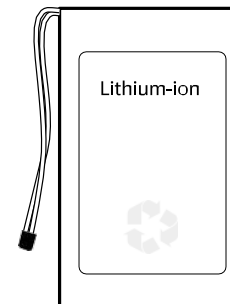
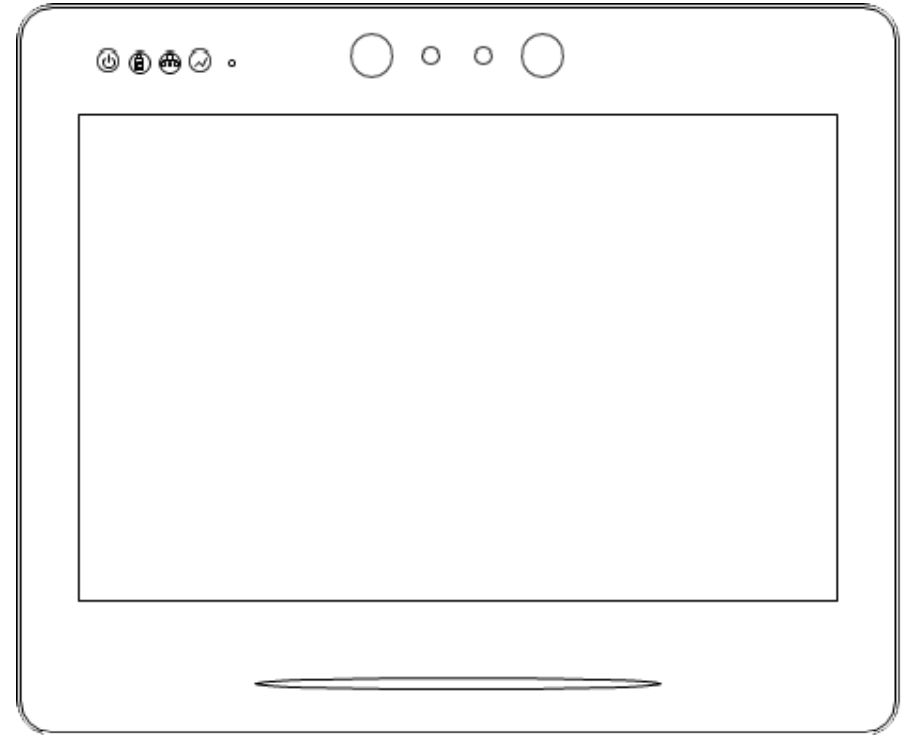
Installing Lithium-ion Battery pack



A Lithium-ion battery pack comes pre-installed on the Ultima 10 but is not connected to the motherboard. Customers who wish to use the battery need to connect it to the motherboard themselves.

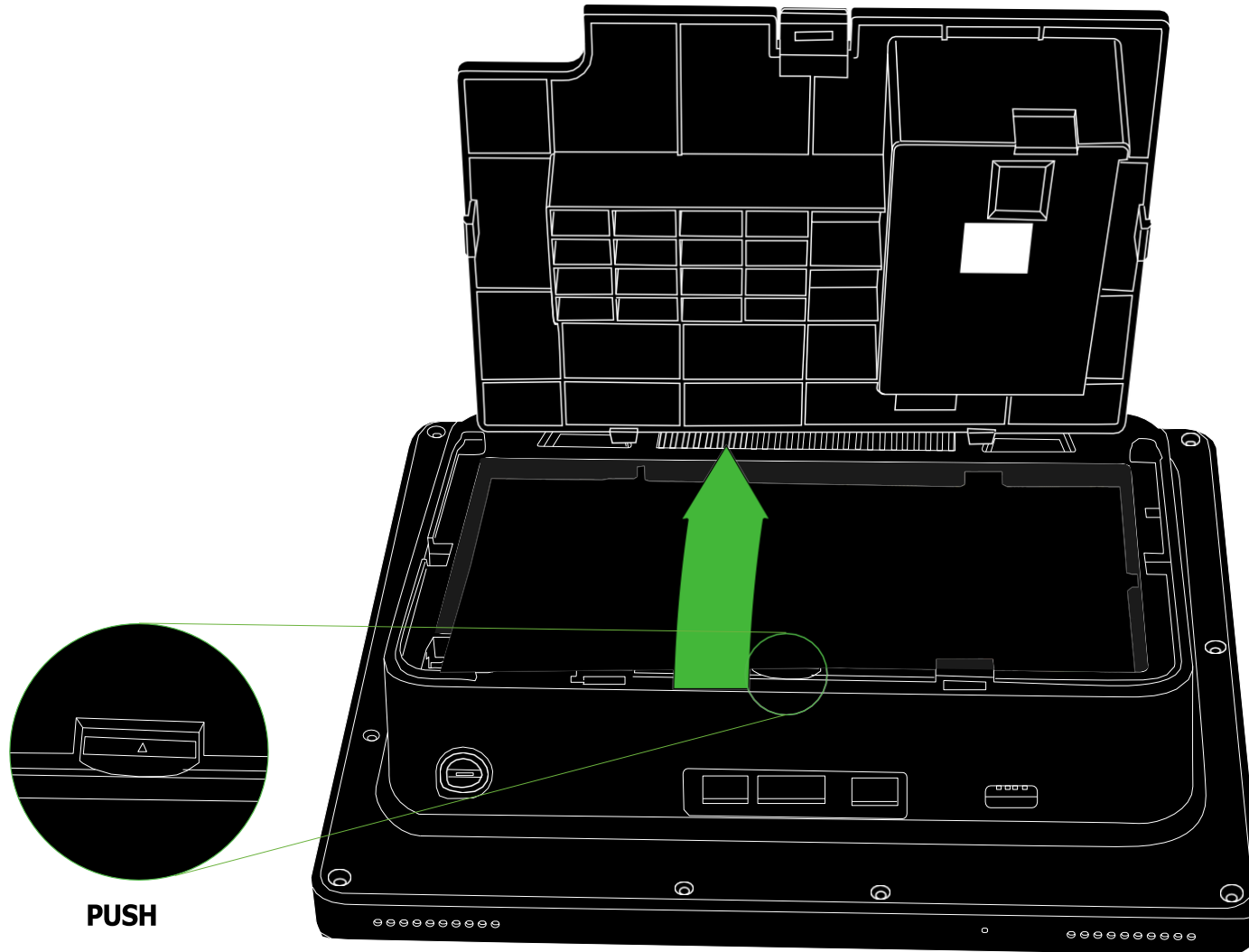
Step 1 : Open the back cover

Step 2 : Connecting wires to motherboard



Lithium-ion Battery

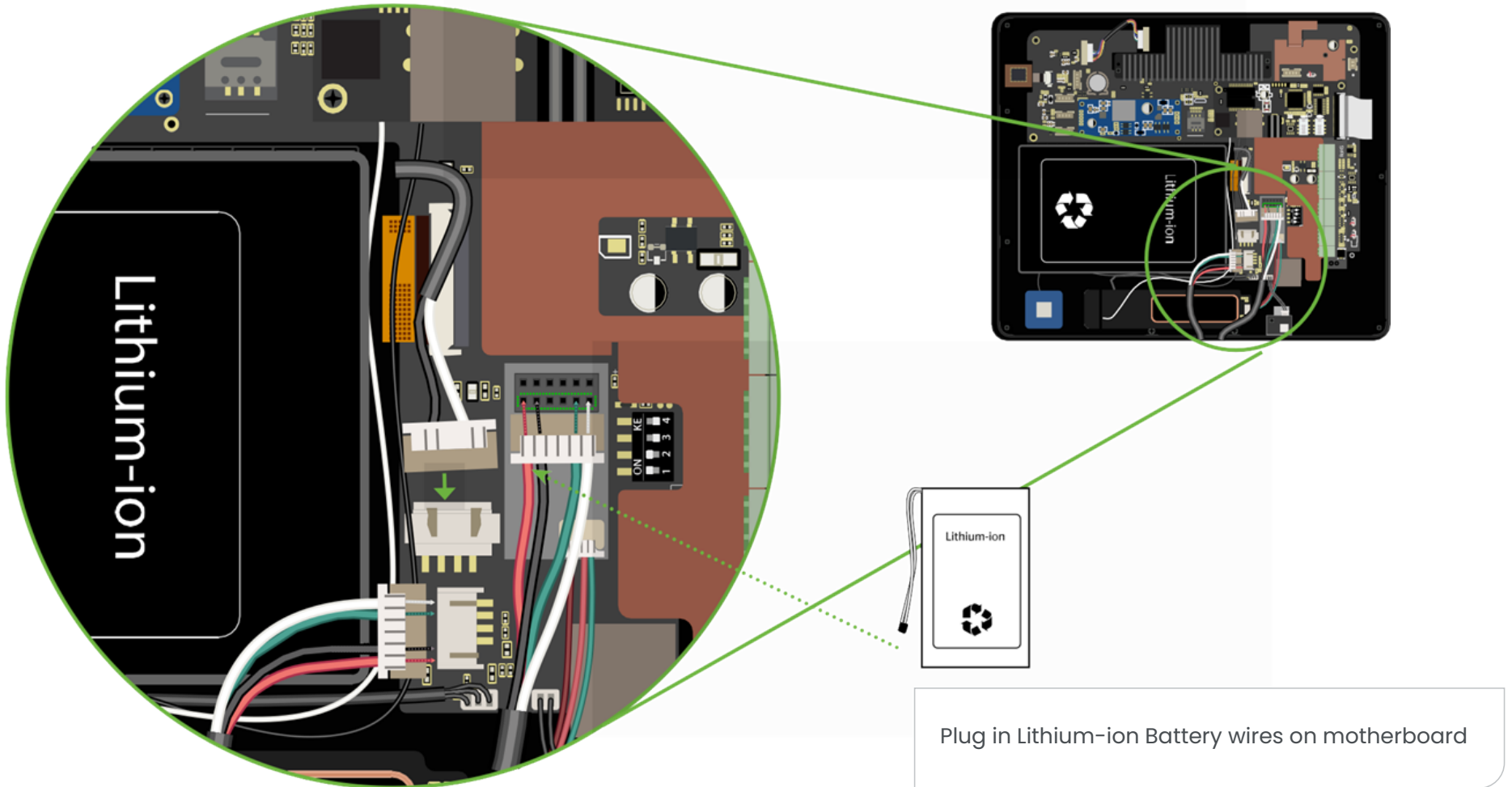
Opening the back cover



Push the latch lock and lift the back cover upward to open assembly completely to access to internal component.

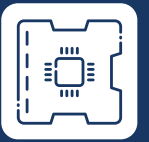
Caution Do not apply force

Installing Lithium-ion Battery | Connecting wires to motherboard

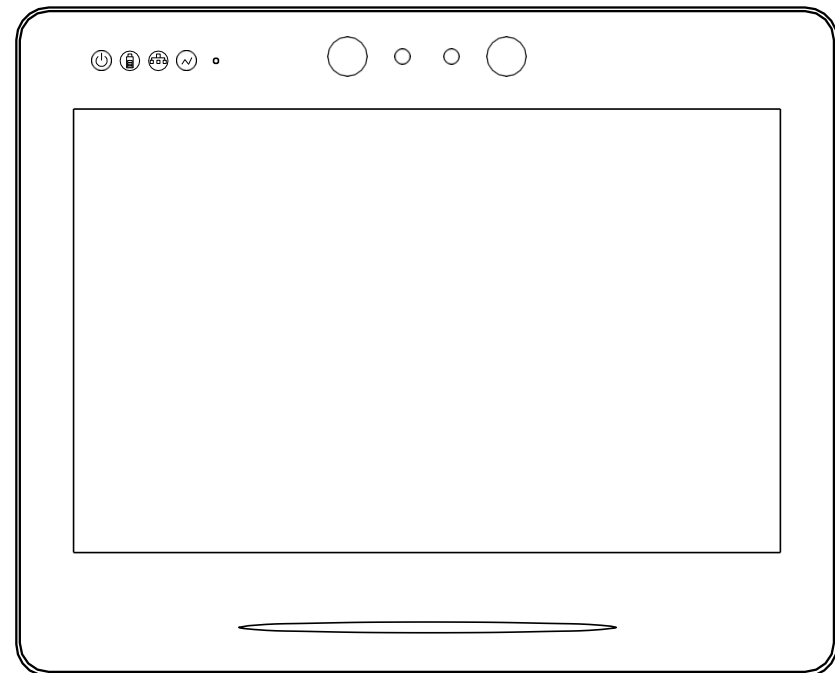




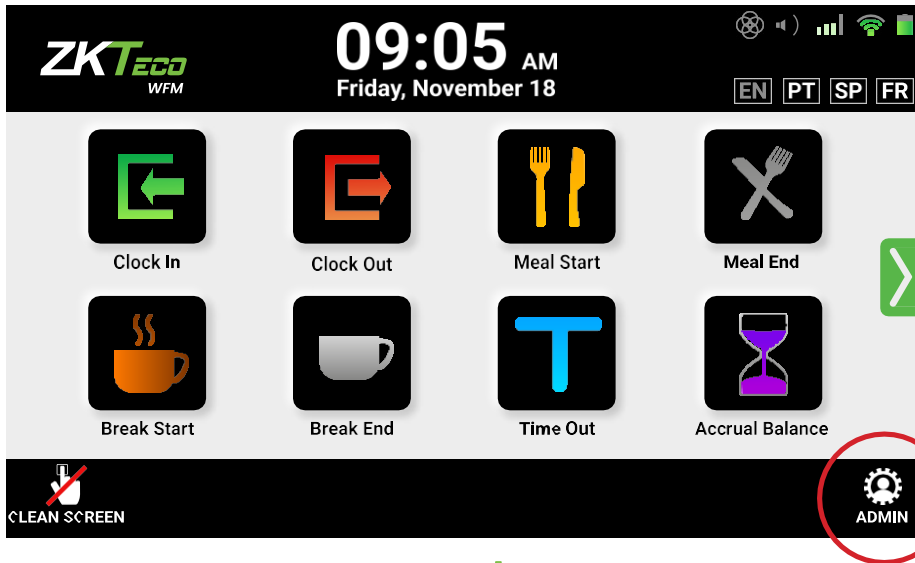
Close the back cover



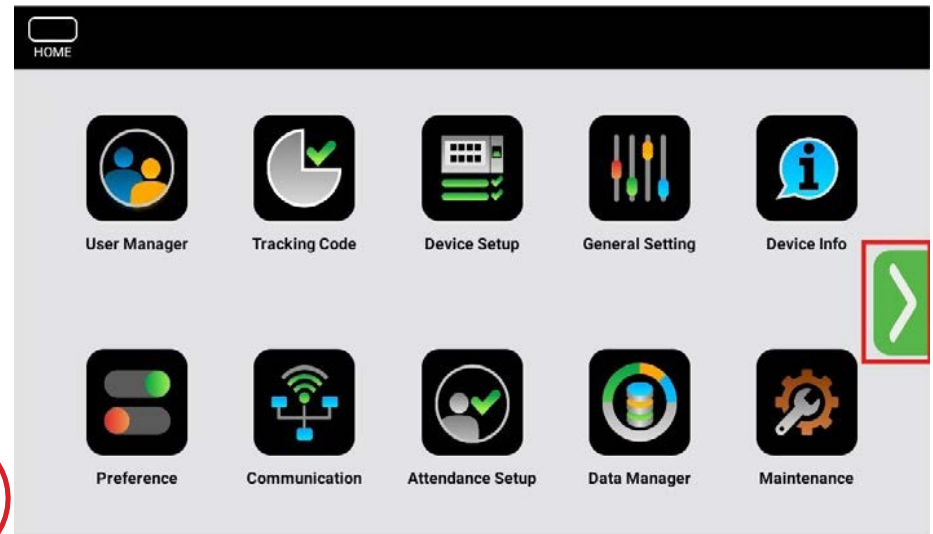
1. Testing Display & Touch
2. Testing Control Relay
3. Testing Voice



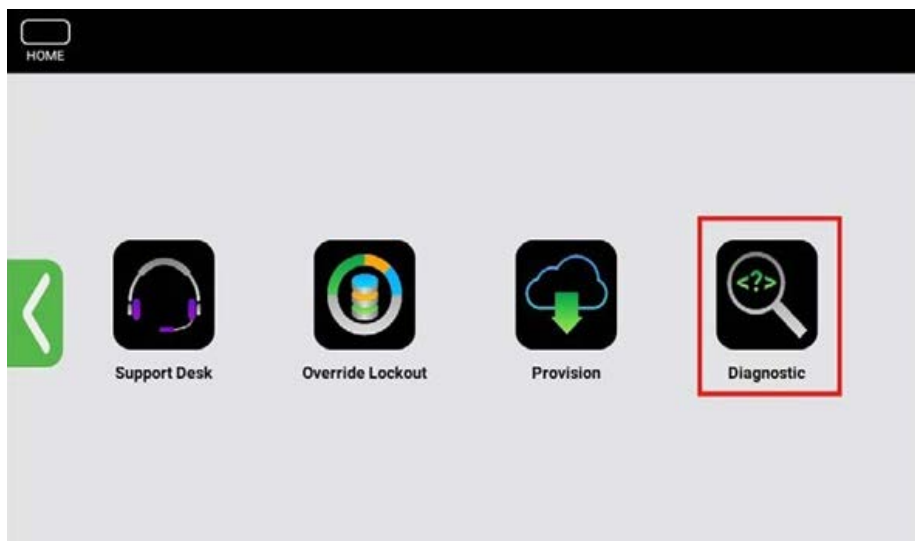
D - Display & Touch | **C** - Control Relay | **V** - Voice



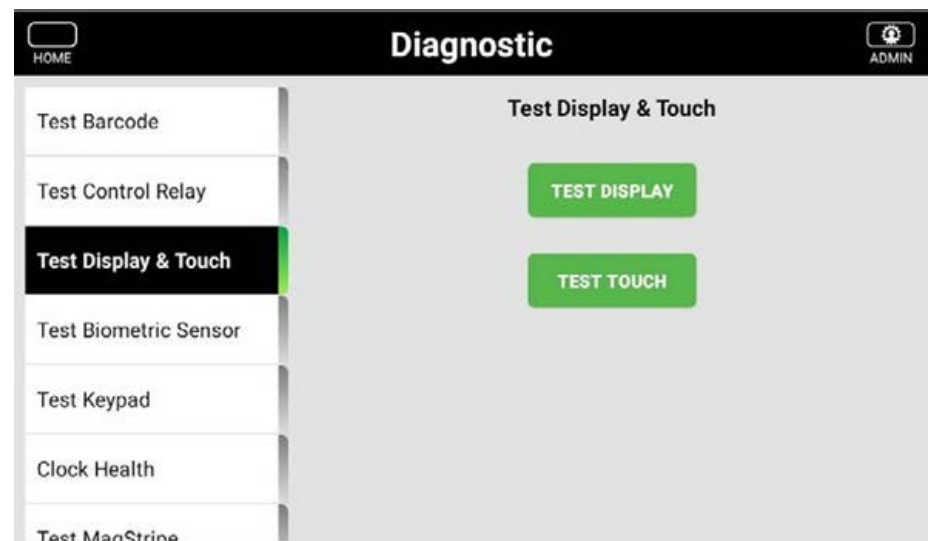
Press Admin  to login



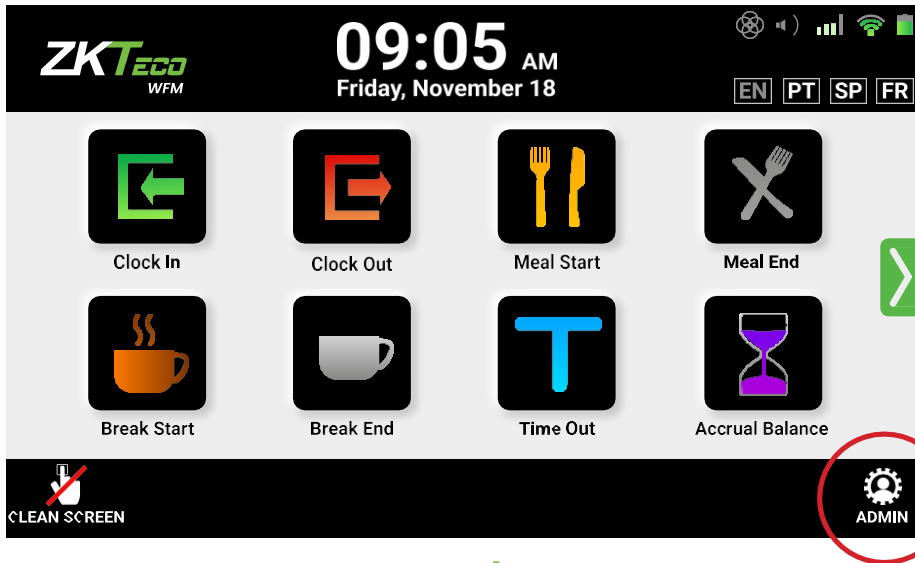
Press next screen >



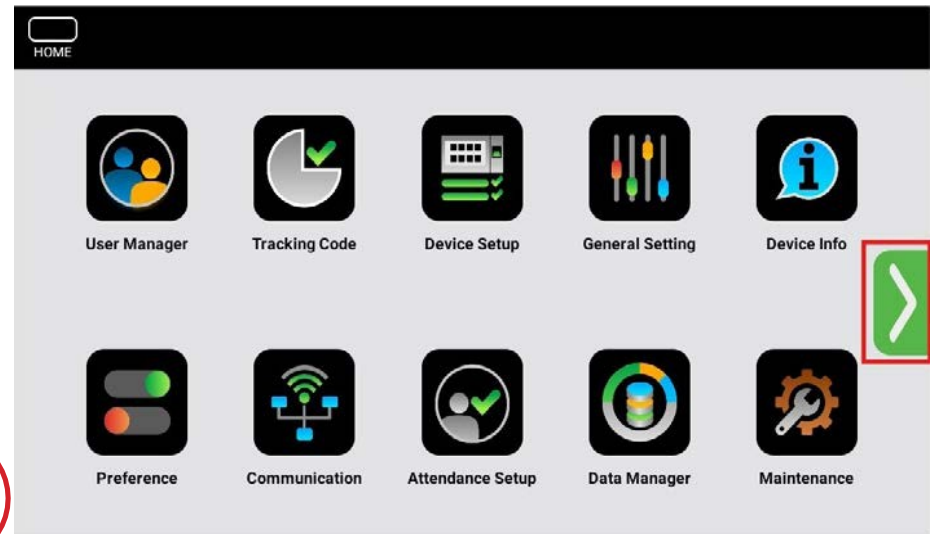
Press Diagnostic



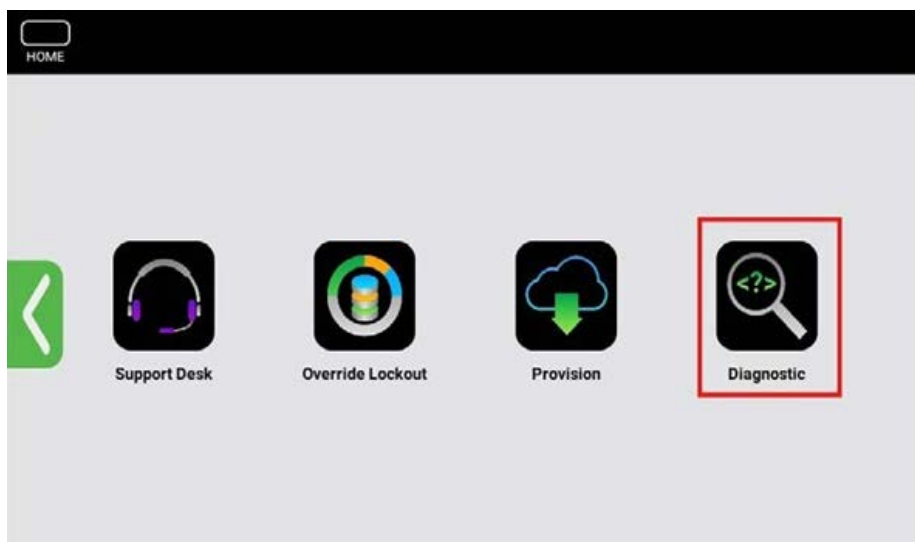
Test Display & Touch



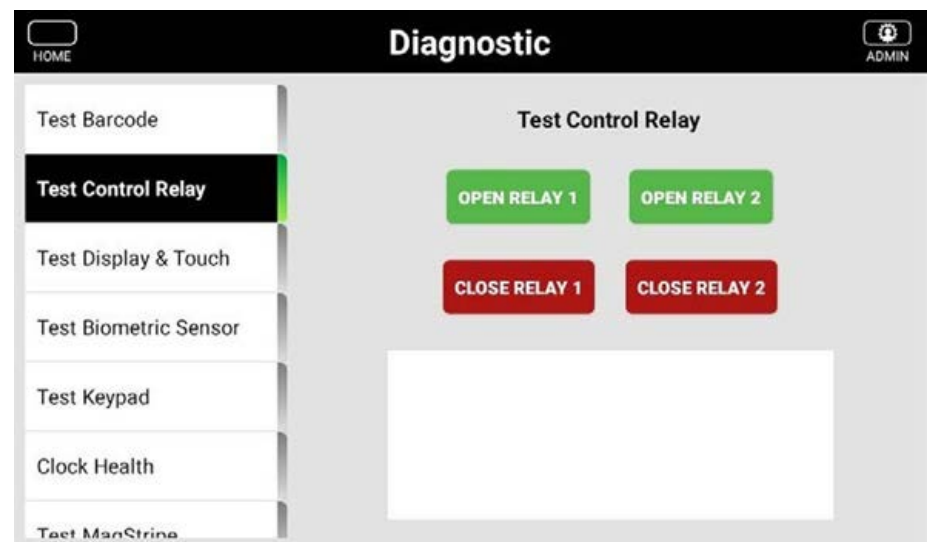
Press Admin to login



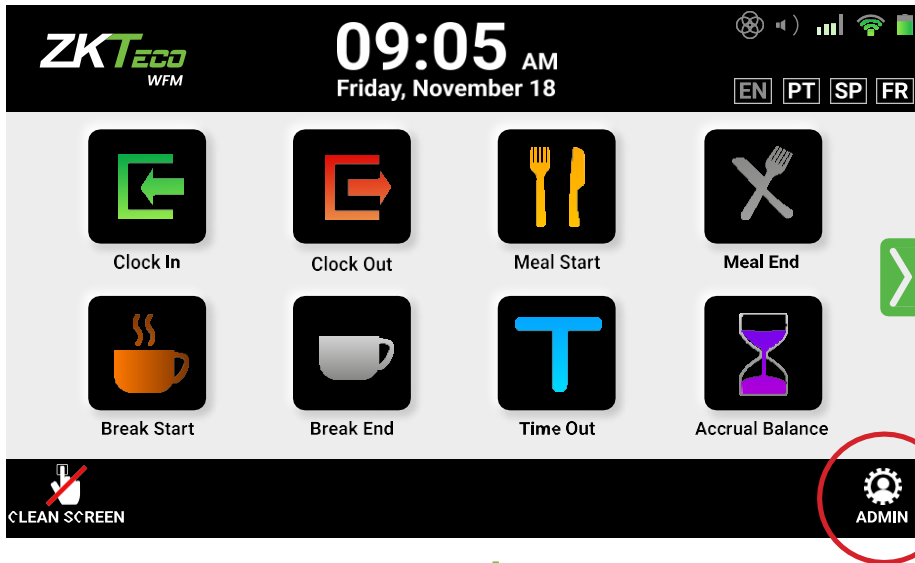
Press next screen



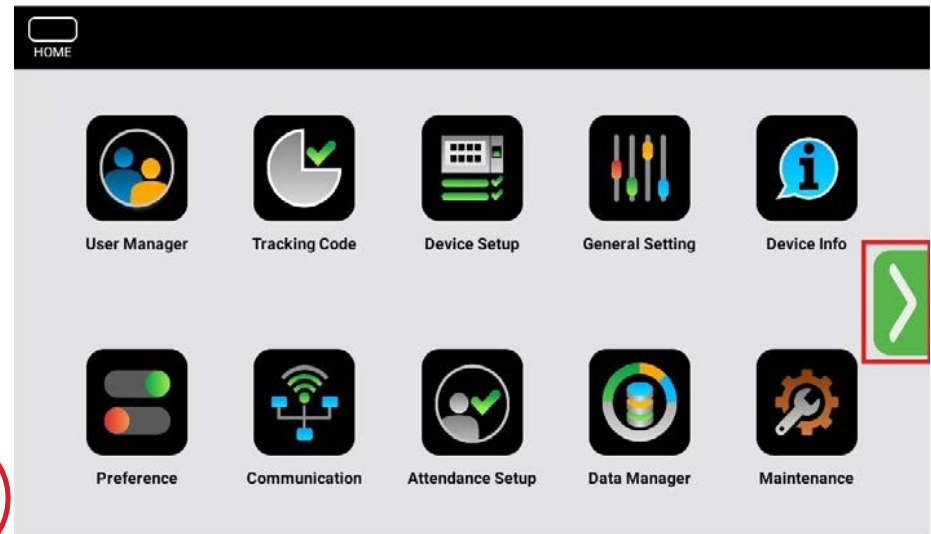
Press Diagnostic



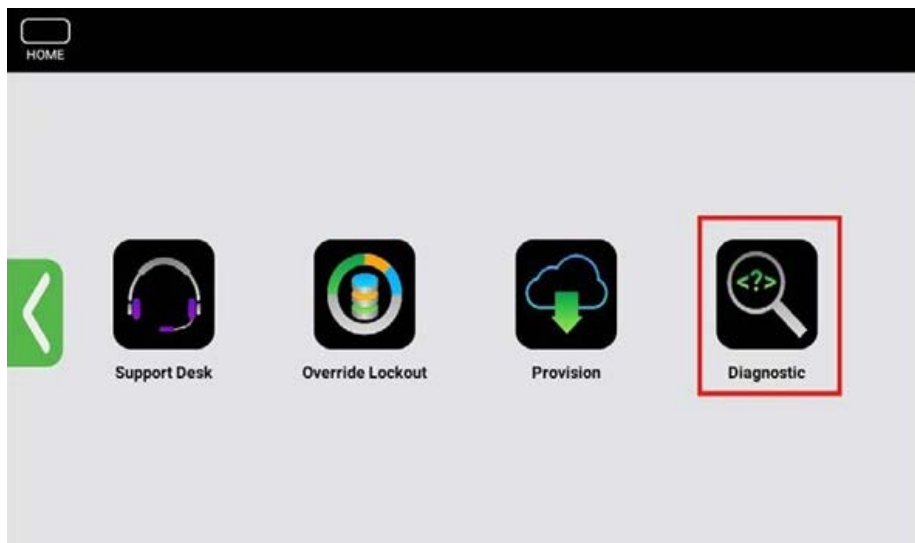
Test Control Relay



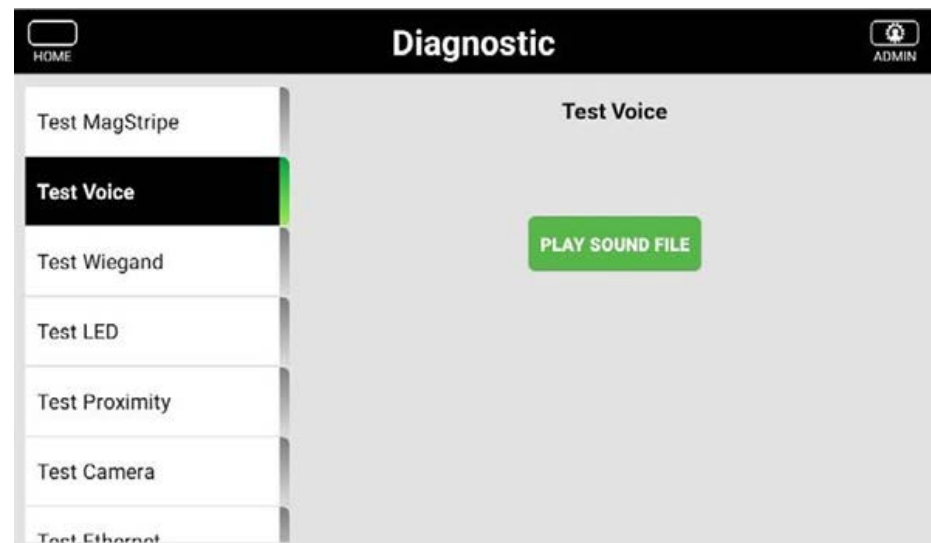
Press Admin to login



Press next screen



Press Diagnostic



Test Voice



Issue	Probable Causes	Corrective Action
There is no power in the Ultima 10	If the Ultima 10 is powered using standard AC power cable, the power supply is not properly plugged into the main board, the AC outlet, or both	Check the connections to the main board and the AC outlet Note: Do not plug the power cord into the AC outlet until you ensure that the power supply is properly connected to the main board
	If the Ultima 10 is powered using Ethernet cable, the network may be down, or connections to and from the PoE+ module may not be secure	• Ensure that the network is running • Ensure that the network cable is connected to the main board
The power LED is turned on, but nothing appears on the screen	If the Ultima 10 is powered using a standard AC power cable, the AC line is not live or is supplying improper voltage	Measure voltage at the AC outlet and, if necessary, locate another power source
	If the Ultima 10 is powered using Ethernet cable, there is an issue with the network	• Check the Ethernet connection at the Ultima 10, and all other network connections • Ensure that the network is supplied with an 802.11af compliant power source
Communication fails	Communication settings are incorrect either at the Ultima 10 or the host PC	Communication settings are incorrect either at the Ultima 10 or the host PC
	Communications cable is connected incorrectly	Ensure that all communications cables are routed correctly and connected properly



Part I

Warning about battery

- Replacement of a battery with an incorrect type that can defeat a safeguard. Therefore, please make sure that the device uses alkaline button cell batteries
- Disposal of a battery into fire or a hot oven, or mechanically crushing or cutting of a battery, that can result in an explosion
- A battery subjected to extremely low air pressure that may result in an explosion or the leakage of flammable liquid or gas

Part II

- Ultima 10 User Manual must show all Wi-Fi Frequency Bands and Maximum-Output-Power. The data as below.

For EU compliance

Wi-Fi Frequency Bands

- Wi-Fi 2.4G: 2412–2472MHz, **Max. output power:** 18.44dBm+
- Wi-Fi 5G: 5180– 5240MHz, 5260–5320MHz, 5500–5700MHz, **Max. output power:** 18.99dBm+.

And also, added restrictions on the use of indoor warnings, as below: 5150–5350MHz are restricted in indoor used for all EU member states

Part III

Hereby, ZKTECO WFM declares that this Product is in compliance with the essential requirements and other relevant provisions of Directive 2014/53/EU

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

1. This device may not cause harmful interference, and
2. This device must accept any interference received, including interference that may cause undesired operation

Warning: Changes or modifications to this unit not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

Note: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules.

These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation.

If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna
- Increase the separation between the equipment and receiver
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected
- Consult the dealer or an experienced radio/TV technician for help

"This equipment complies with FCC RF radiation exposure limits set forth for an uncontrolled environment. This equipment should be installed and operated with a minimum distance of 20 cm between the radiator and your body. This transmitter must not be co-located or operating in conjunction with any other antenna or transmitter".



Fingerprint Enrollment
Matching Information



Datasheet



Privacy Concerns

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